

Complete CSS

Course Notes

Contains 1 PDF lessons

Generated: 6 April 2026

KnowledgeGate

Module

CSS

Topic

CSS

Subtopic

CSS

Lesson

CSS Notes

Lesson 1 of 1

COMPLETE



CSS

7 HOUR



**MYNTRA
PROJECT**



CERTIFICATE

NOTES



Ex- amazon  Microsoft



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What is IDE

1. IDE stands for Integrated Development Environment.
2. Software suite that consolidates basic tools required for software development.
3. Central hub for coding, finding problems, and testing.
4. Designed to improve developer efficiency.



Need of IDE

1. **Streamlines** development.
2. Increases **productivity**.
3. **Simplifies** complex tasks.
4. Offers a **unified** workspace.
5. **IDE** Features
 1. Code **Autocomplete**
 2. Syntax **Highlighting**
 3. **Version Control**
 4. **Error Checking**



```
MainActivity.java

@Composable
fun MessageCard(msg: Message) {
    Row(modifier = Modifier.padding(8.dp)) {
        Image(
            painter = painterResource(R.drawable.android_studio_logo),
            contentDescription = "Profile Picture",
            modifier = Modifier
                .size(45.dp)
        )

        Spacer(modifier = Modifier.width(8.dp))
        Column(modifier =
            background(Color = Color.White)) {
            Text(text = msg.author, color = Color.Black)
            Spacer(modifier = Modifier.height(1.dp))
            Text(text = msg.body, color = Color.Black)
        }
    }
}
```



Installation & Setup

1. Search VS Code

[KNOWLEDGEGATE]



VsCode Extensions

1. Live Server
2. Prettier



FrontEnd / BackEnd / FullStack



Client Side / Front-
End
Web Development

Server Side
Back-End



Full Stack

Role of Browser



1. **Displays Web Page:** Turns HTML code into what you see on screen.
2. **User Clicks:** Helps you interact with the web page.
3. **Updates Content:** Allows changes to the page using JavaScript.
4. **Loads Files:** Gets HTML, images, etc., from the server.



HTML

(Hypertext Markup Language)

1. **Structure:** Sets up the layout.
2. **Content:** Adds **text**, images, links.
3. **Tags:** Uses elements like `<p>`, `<a>`.
4. **Hierarchy:** Organizes elements in a tree.





CSS

(Cascading Style Sheets)

1. **Style:** Sets the look and feel.
2. **Colors & Fonts:** Customizes text and background.
3. **Layout:** Controls position and size.
4. **Selectors:** Targets specific **HTML** elements.





JS

(Java Script)

1. JavaScript has nothing to do with Java
2. **Actions:** Enables interactivity.
3. **Updates:** Alters page without reloading.
4. **Events:** **Responds** to user actions.
5. **Data:** Fetches and sends info to server.





Importance of CSS



Premium Website



Premium Brand



Premium Customer



High Salary Developer

 HTML required for CSS

COMPLETE

 HTML

4 HOUR

PROJECT

CERTIFICATE

CODE



NOTES



HTML Revision Covered

Course Levels



Level 1 CSS Basics

Level 2 Color System and Background

Level 3 Text Properties

Level 4 Box Model CSS

Level 5 Display & Position

Level 6 Flexbox, Grid & Media Queries

Level 7 Animation, Transition & Transform

Project:  Myntra Clone

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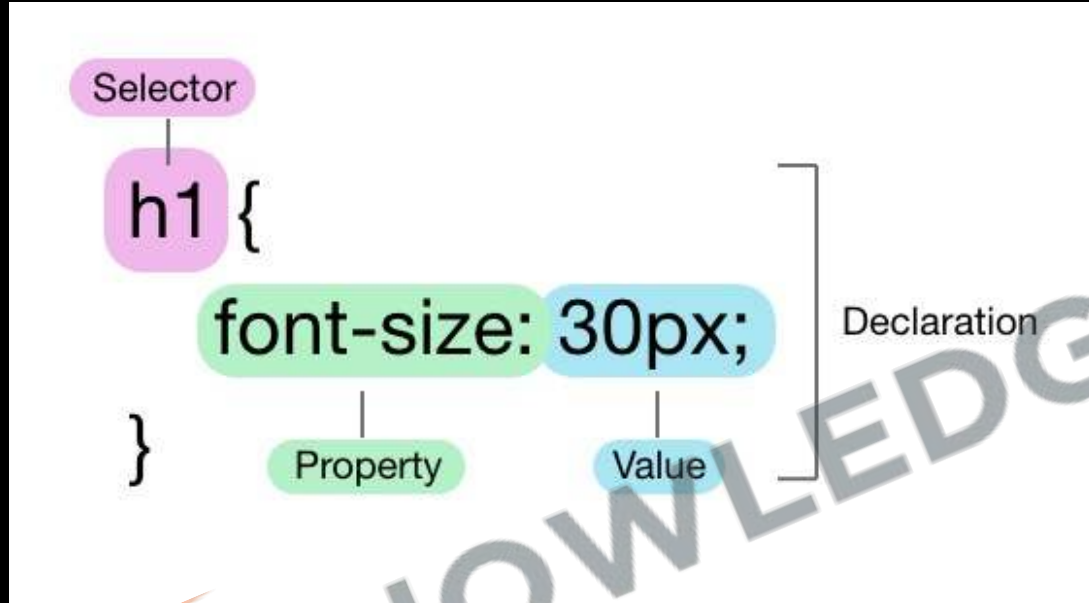
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Level 1

CSS Basics

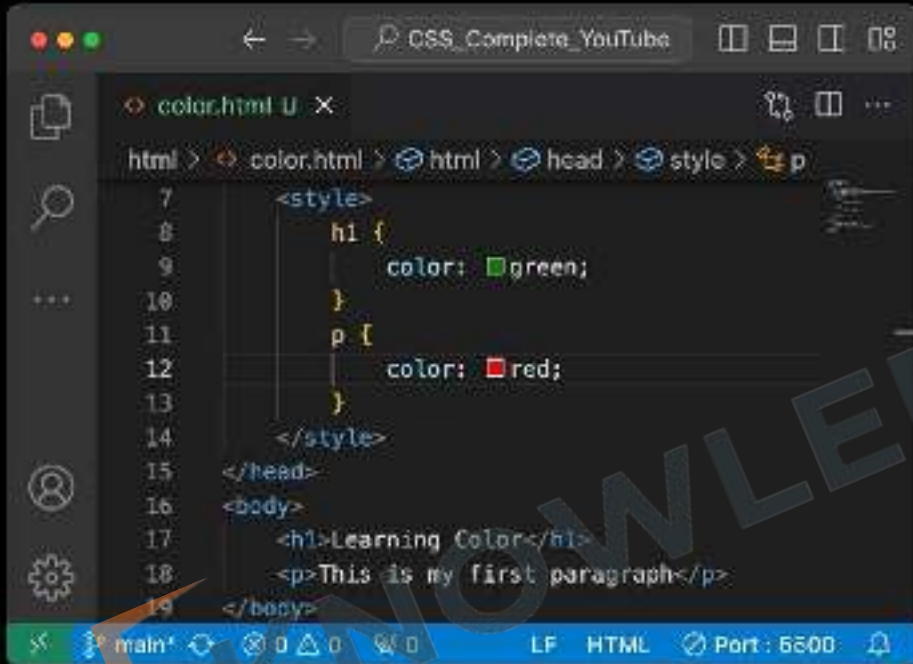
1. Basic Syntax
2. Color Property
3. Including Styles
4. HTML Refresher
5. Comments
6. MDN Documentation
7. Browser Tools
8. Selectors

1. Basic Syntax



- **Selector:** The HTML element that you want to style.
- **Property:** The attribute you want to change (like font, color, etc.).
- **Value:** The specific style you want to apply to the property (like red, bold, etc.).

2. Color Property



```
color.html U X
html > color.html > html > head > style > p
7   <style>
8     h1 {
9       color: green;
10    }
11    p {
12      color: red;
13    }
14  </style>
15 </head>
16 <body>
17   <h1>Learning Color</h1>
18   <p>This is my first paragraph</p>
19 </body>
```



- **Definition:** The CSS color property defines the text color or foreground color in an HTML element.
- **Enhancement:** Use it to emphasize sections and elevate webpage aesthetics.

3. Including Styles (Inline Styling)

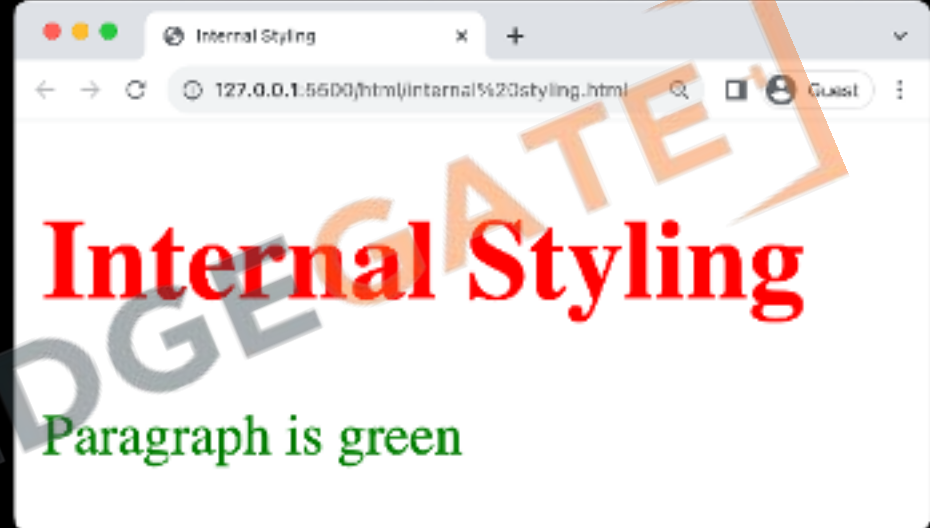
```
html > <> inline styling.html > html
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4    <title>Inline Styling</title>
5  </head>
6  <body>
7    <h1 style="color: red;">Inline Styling</h1>
8    <p style="color: green;">Paragraph is green</p>
9  </body>
10 </html>
```



- **Direct Application:** Apply styles directly to **HTML** elements using the style attribute.
- **One-off Changes:** Ideal for **single**, unique style alterations.
- **Can Be Cluttered:** May lead to **cluttered** HTML if used extensively.
- **Limited Reusability:** Reduces the **reusability** of CSS rules in larger projects.

3. Including Styles (Internal Styling)

```
3 <head>
4   <title>Internal Styling</title>
5   <style>
6     h1 {color: red;}
7     p {color: green;}
8   </style>
9 </head>
10 <body>
11   <h1>Internal Styling</h1>
12   <p>Paragraph is green</p>
13 </body>
14 </html>
```



- **Embedded CSS:** Styles are placed within `<style>` tags in the HTML head section.
- **Cleaner than Inline:** More organized compared to inline styles.
- **Reusable Styles:** Allows for some reuse of styles across the page.

3. Including **Styles** (External Styling)

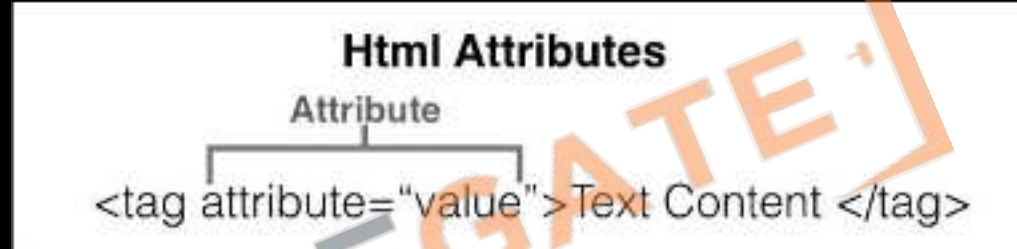
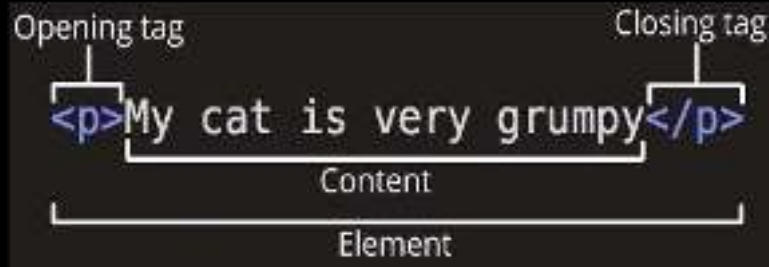
```
3 <head>
4   <title>External Styling</title>
5   <link rel="stylesheet" href="../css/
6     external styling.css">
7 </head>
8 <body>
9   <h1>External Styling</h1>
10  <p>Paragraph is green</p>
11 </body>
```

```
css > # external styling.css > p
1  h1 {
2    color: red;
3  }
4
5  p {
6    color: green;
7  }
```



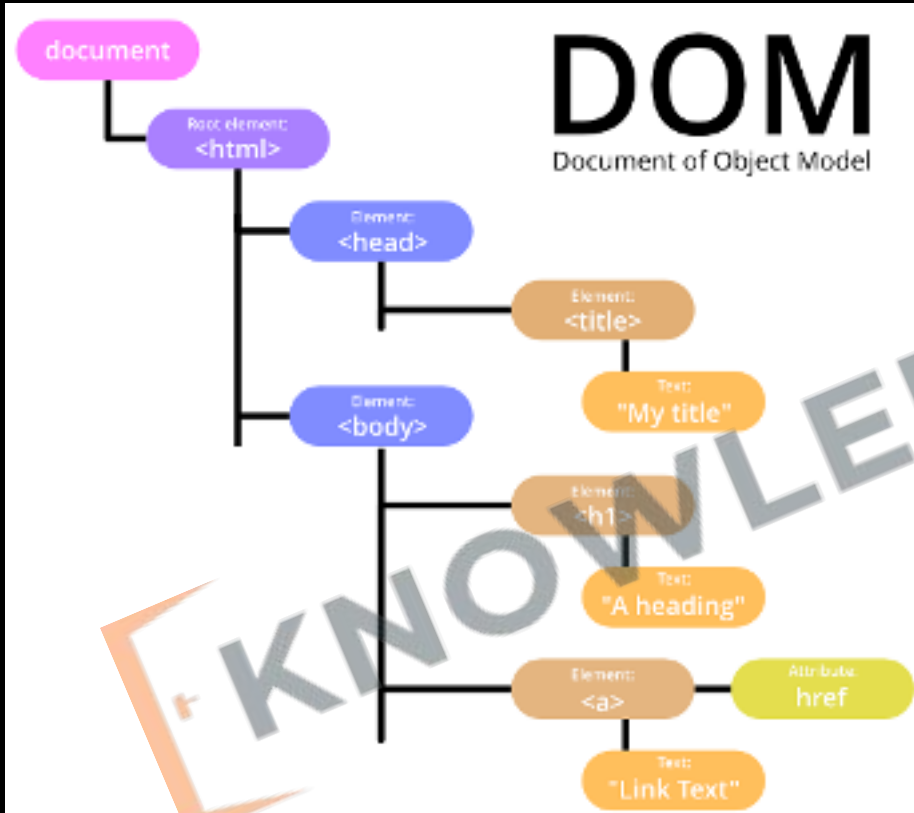
- **Separate CSS File:** Stores styles in a **separate .css file**, linked to HTML.
- **Reusable:** Enables style **reuse** across multiple webpages.
- **Link in HTML:** Use the **<link>** tag within the **<head>** section to link the CSS.
- **Relative or Absolute Path:** The href attribute can contain a relative or absolute path to the CSS file.

4. HTML Refresher (Tags & Attributes)



1. **Elements** that are used to create a website are called HTML Tags.
 2. Tags can contain **content** or **other HTML tags**.
 3. Define elements like **text**, **images**, **links**
1. Provides **additional information** about elements
 2. Use **name=value** format

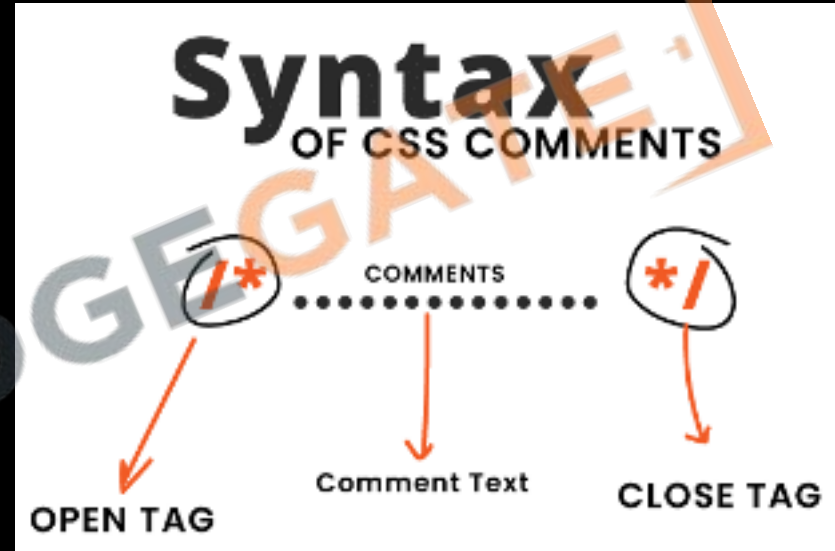
4. HTML Refresher (DOM)



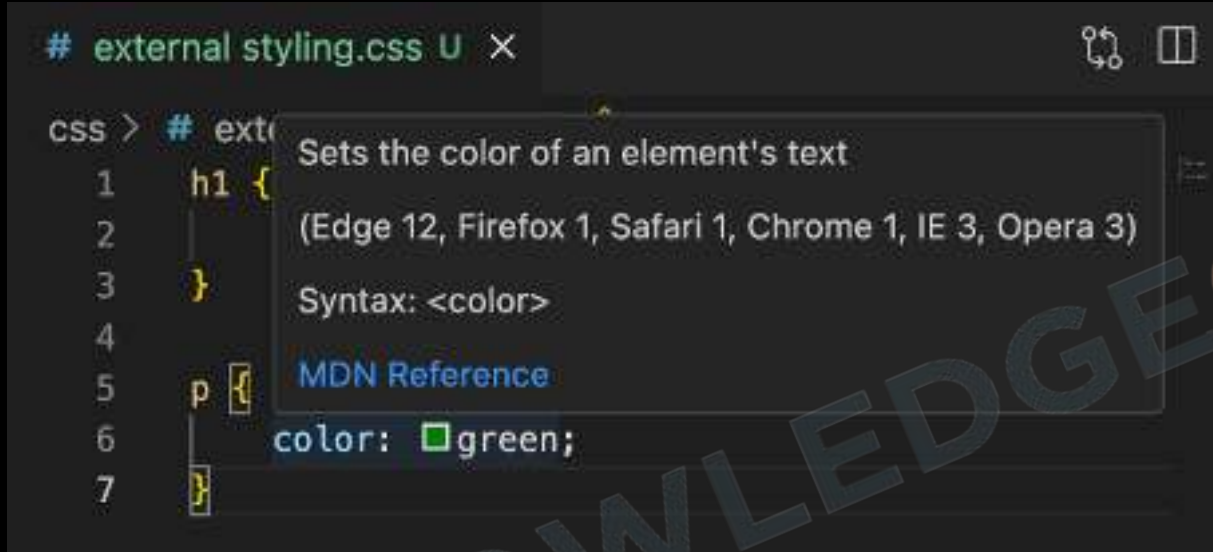
- 1. Structure Understanding:** Helps in understanding the **hierarchical structure** of a webpage, crucial for applying targeted CSS styles.
- 2. Dynamic Styling:** Enables learning about dynamic styling, allowing for **real-time changes** and interactivity through CSS.

5 Comments

1. Used to add **notes** in HTML or CSS code
2. **Not displayed** on the web page
3. Syntax: `<!-- Comment here -->`
4. Helpful for **code organization**
5. Can be **multi-line** or **single-line**

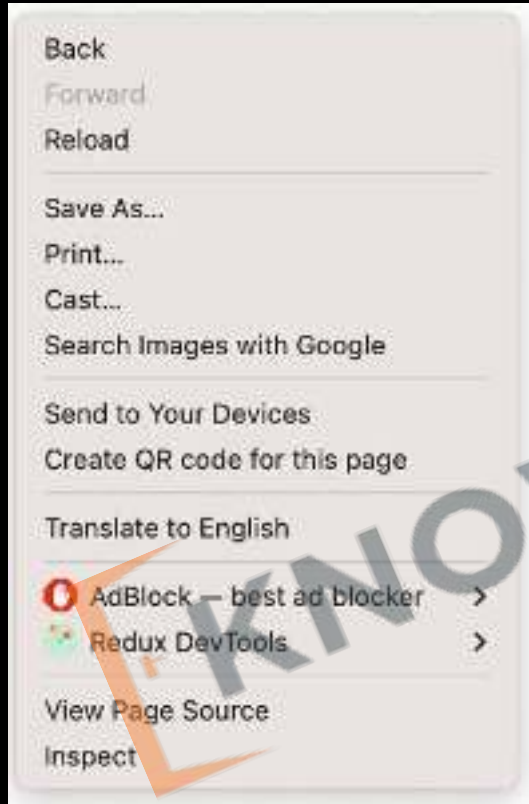


6.MDN Documentation



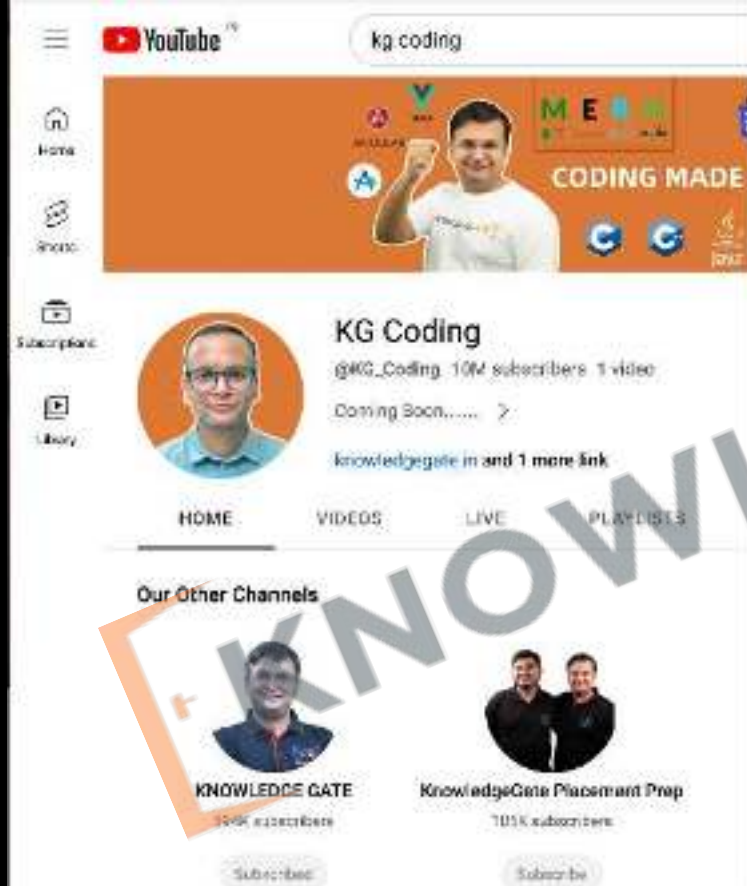
1. For Official resource for CSS, visit <https://developer.mozilla.org/>
2. Includes examples for real-world use
3. Updated with latest CSS-3 features
4. Trusted by developers worldwide

7. Browser Tools (Inspect element)



1. Allows **real-time editing** of HTML/CSS
2. Useful for debugging and testing
3. Shows **element hierarchy** and layout
4. **Highlights** selected elements on page

7. Browser Tools (HTML Live Edit)



Changed **Subscriber** count

7. Browser Tools (CSS Live Edit)

Changed **Channel**
Name color



7. Browser Tools (JS Live Edit)

The image displays a web browser window with a custom interface and its developer tools open.

Browser Interface:

- Header:** YouTube logo, Search bar.
- Left Sidebar:** Home, Shorts, Subscriptions, Library.
- Main Content:** A video player area showing a man in a white t-shirt with "CODING MAD" text. Above him are logos for Angular, Vue.js, and MERN stack.
- Bottom Navigation:** HOME, VIDEOS, LIVE, PLAYLISTS.

Developer Tools (Console):

- Errors:** Two "DevTools failed to load source map" errors are visible, both with status code 404.
- Log:** A log entry shows a GET request to `https://www.google.com/pagead/...` with status 200.
- Network:** A list of network requests is shown, including `https://www.google.com/pagead/...` and `https://static.devtools.io/...`.
- JavaScript:** A snippet of JavaScript code is visible, showing a function call to `document.getElementById("inner-header-container").style.visibility = "hidden"`.

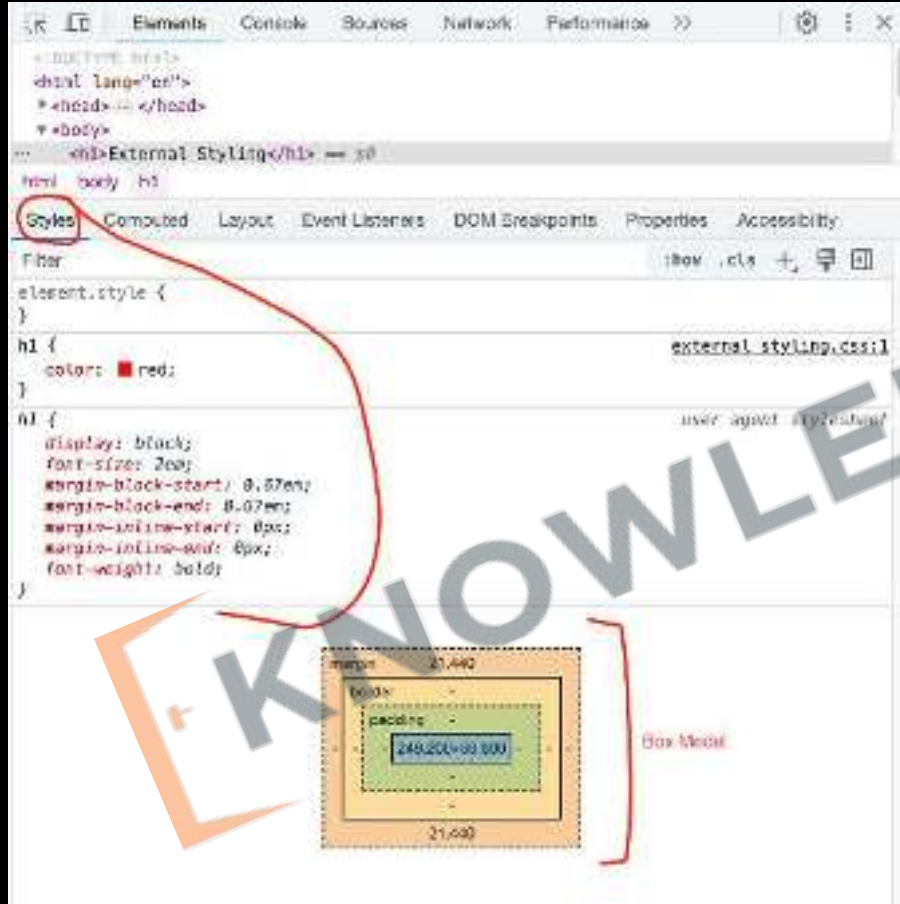
7. Browser Tools (Changes happening at Client)

1. Changes made are temporary
2. Affect only the current session
3. Not saved to the server
4. Reset upon page reload
5. Useful for testing, not permanent fixes

Like: If you change the question in your question paper that has no effect on actual exam.



7. Browser Tools (CSS Specific element)



- Styles Panel
- Box Model
- Changes happening only at client

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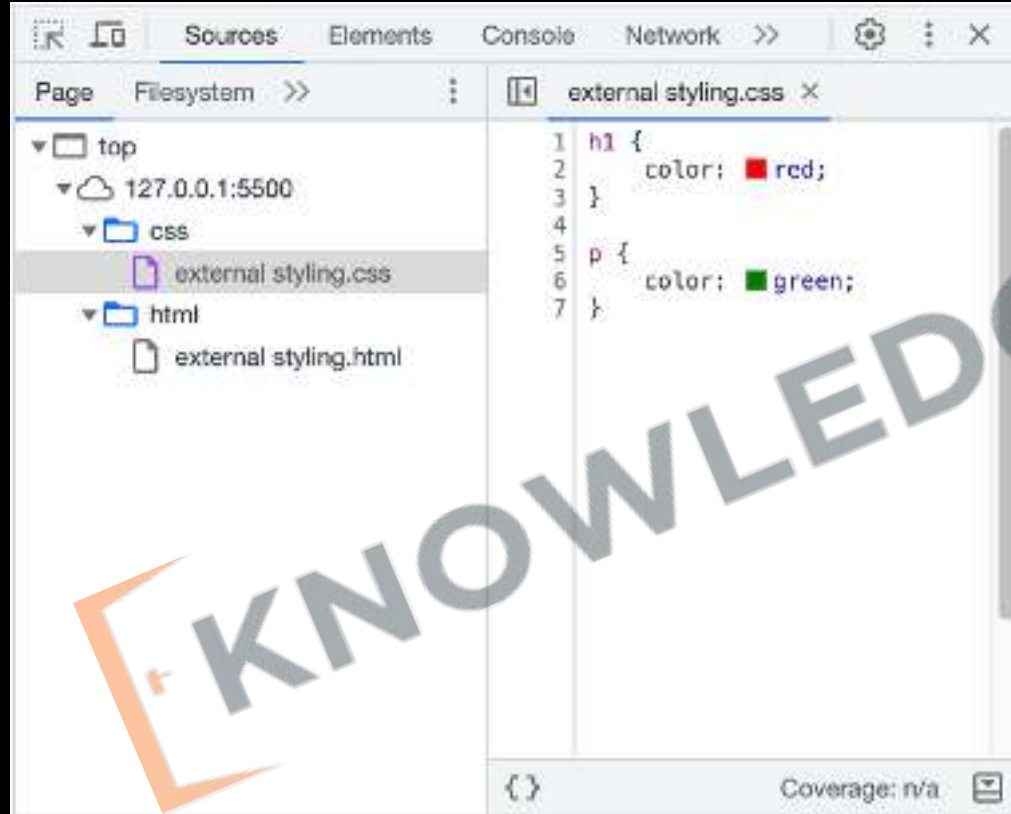


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7. Browser Tools (Source tab)

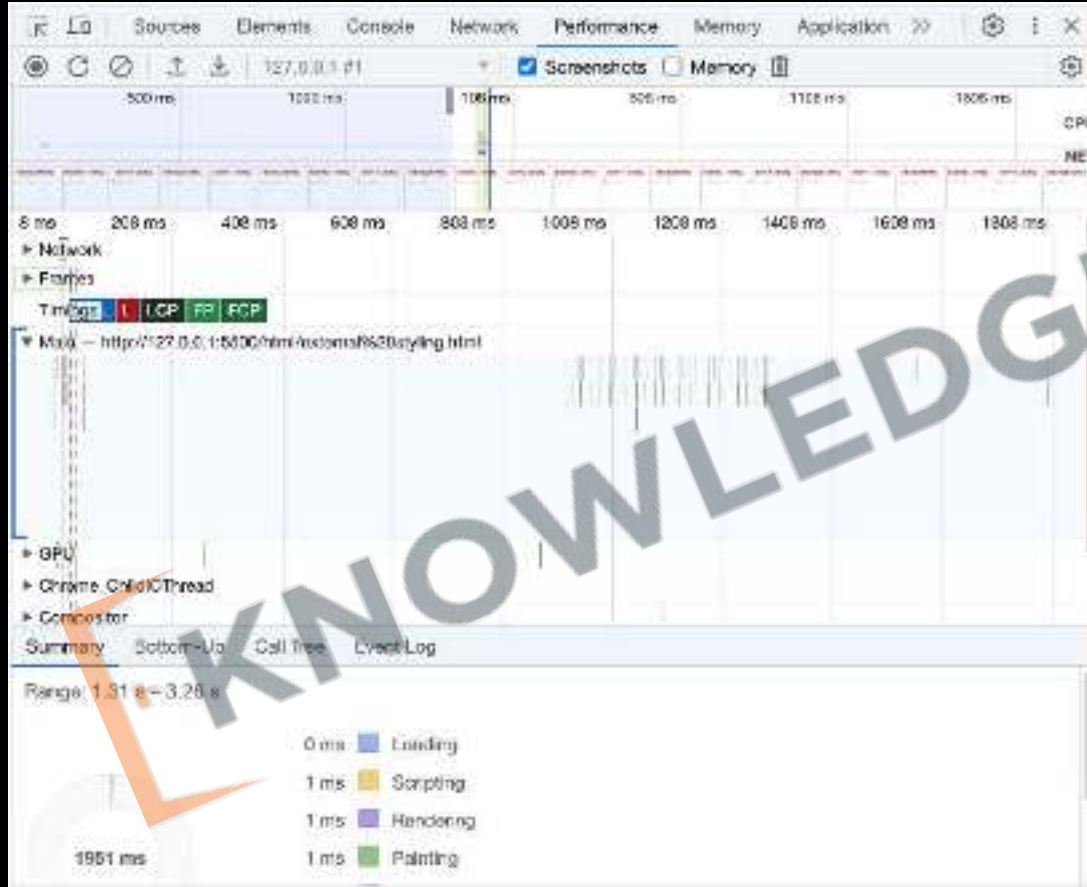


7. Browser Tools (Network tab)

The screenshot shows the Chrome DevTools Network tab. The top bar includes tabs for Sources, Elements, Console, Network (selected), Performance, Memory, and Application. Below the tabs, there are icons for recording, pausing, and filtering, along with checkboxes for 'Preserve log', 'Disable cache', and 'No throttling'. A search filter is set to 'Filter'. Below the filter, there are checkboxes for 'Invert' and 'Hide data URLs'. A list of request types is shown: All, Fetch/XHR, JS, CSS, Img, Media, Font, Doc, WS, Wasm, Manifest, Other, and 'Has blocked cookies'. Below this, there are checkboxes for 'Blocked Requests' and '3rd-party requests'. A timeline view shows a request at approximately 10 ms. Below the timeline, a table lists the network requests.

Name	Status	Type	Initiator	Size	Time	Waterfall
external%20styling.html	304	docu...	Other	295 B	4 ms	
ws	101	webso...	external stylin...	0 B	Pending	

7. Browser Tools (Performance tab)



8. Selectors (Element selector)

```
3 <head>
4   <title>Element Selector</title>
5   <style>
6     h1 {
7       color: red
8     }
9   </style>
10 </head>
11 <body>
12   <h1>Universal Selector</h1>
13   <p>Lorem ipsum dolor sit amet consectetur
14     adipiscing elit. Nulla, nisi.</p>
```



- **Targets Elements:** Selects HTML elements based on their **tag name**.
- **Syntax:** Simply use the **element's name**
- **Uniform Styling:** Helps in applying **consistent styles** to all instances.
- **Ease of Use:** Straightforward and **easy** to implement for basic styling.

8. Selectors (Universal selector)

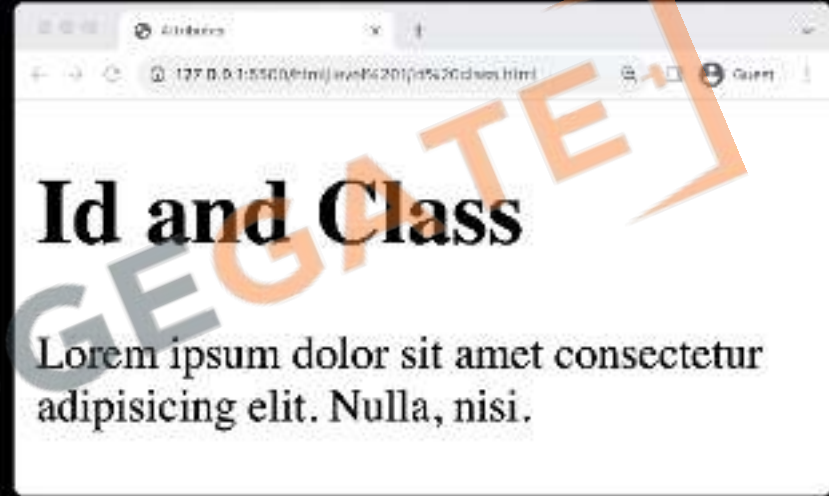
```
3 <head>
4   <title>Universal Selector</title>
5   <style>
6     * {
7       color: red
8     }
9   </style>
10 </head>
11 <body>
12   <h1>Universal Selector</h1>
13   <p>Lorem ipsum dolor sit amet consectetur
14     adipiscing elit. Nulla, nisi.</p>
15 </body>
```



- **Matches All:** Targets and styles **all elements** on a webpage.
- **Syntax:** Utilized as an **asterisk (*)**.
- **Resets Styles:** Commonly used to **reset margins** and paddings globally.
- **Broad Styling:** Useful for setting universal attributes like **font or color**.
- **Usage Caution:** Can cause style **conflicts** due to its wide-reaching effects.

8. Selectors (id & class property)

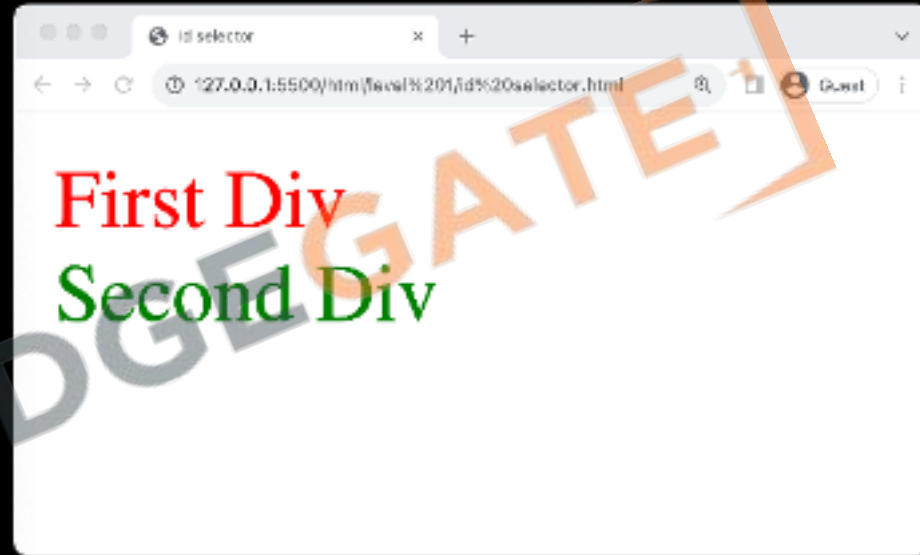
```
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <title>Attributes</title>
5 </head>
6 <body>
7   <h1 id="top_heading">Id and Class</h1>
8   <p class="article">Lorem ipsum dolor sit amet
9     consectetur adipisicing elit. Nulla, nisi.</p>
10 </body>
</html>
```



- **ID Property:** Assigns a unique identifier to a single HTML element.
- **Class Property:** Allows grouping of multiple HTML elements to style them collectively.
- **Reusable Classes:** Class properties can be reused across different elements for consistent styling.
- **Specificity and Targeting:** Both properties assist in targeting specific elements or groups of elements for precise styling.

8. Selectors (Id selector)

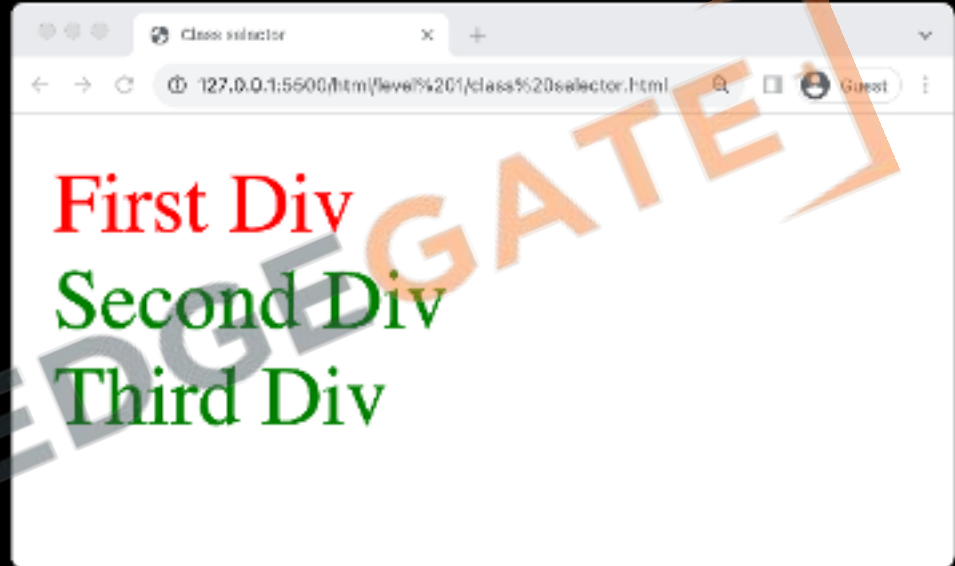
```
3 <head>
4   <title>Id selector</title>
5   <style>
6     #first { color: red; }
7     #second { color: green; }
8   </style>
9 </head>
10 <body>
11   <div id="first">First Div</div>
12   <div id="second">Second Div</div>
13 </body>
```



- **Unique Identifier:** Targets a specific element with a unique ID attribute.
- **Syntax:** Uses the hash (#) symbol
- **Single Use:** Each ID should be used once per page for uniqueness.
- **Specific Targeting:** Ideal for styling individual, distinct elements.

8. Selectors (Class selector)

```
<head>
  <title>Class selector</title>
  <style>
    #first { color: red; }
    .second { color: green; }
  </style>
</head>
<body>
  <div id="first">First Div</div>
  <div class="second">Second Div</div>
  <div class="second">Third Div</div>
</body>
```



- **Group Styling:** Allows styling of multiple elements grouped under a class.
- **Syntax:** Utilizes the dot (.) symbol.
- **Reusable:** Can be used on multiple elements for consistent styling.
- **Versatility:** Ideal for applying styles to a category of elements.

8. Selectors (Group selector)

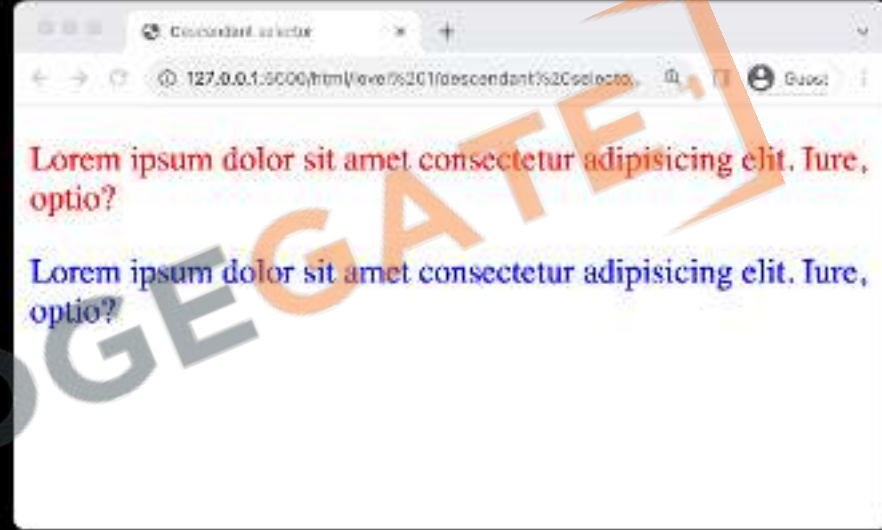
```
<head>
  <title>Group selector</title>
  <style>
    h1, h2, h3 {
      color: red
    }
  </style>
</head>
<body>
  <h1>Heading 1</h1>
  <h2>Heading 2</h2>
  <h3>Heading 3</h3>
  <h4>Heading 4</h4>
</body>
```



- **Multiple Elements:** Styles multiple elements simultaneously.
- **Syntax:** Separates selectors with commas.
- **Efficiency:** Reduces code redundancy and saves time.

8. Selectors (Descendant selector)

```
<head>
  <title>Descendant selector</title>
  <style>
    div p { color: red; }
    p { color: blue; }
  </style>
</head>
<body>
  <div>
    <p>
      Lorem ipsum dolor sit amet consectetur adipisicing
      elit. Iure, optio?
    </p>
  </div>
  <p>
    Lorem ipsum dolor sit amet consectetur adipisicing
    elit. Iure, optio?
  </p>
</body>
```



- **Nested Targeting:** Styles elements **nested** within a specified element.
- **Syntax:** Separate selectors with **spaces**.
- **Hierarchy-Based:** Allows styling based on the **hierarchical** structure of HTML.
- **Specific Styling:** Facilitates more **targeted** and specific styling of elements.

Level 1 Revision

CSS Basics

1. Basic Syntax
2. Color Property
3. Including Styles
4. HTML Refresher
5. Comments
6. MDN Documentation
7. Browser Tools
8. Selectors



Practice Set Level 1

CSS Basics

- Create a heading and set the text color **red**.
- Create a div with id **#heading**, include CSS using all 3 ways **line**, **style tag** and **external**, and observe priority.
- Add **comments** to your **CSS** class
- Create a **div**, **paragraph** and **heading** and use id Selector, element selector and class selector for them.
- Create two divs with id **first** and **second** and define color for both using group selector.



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Level 2

Color System & Background

9. Background-Color Property
10. Color System
11. Absolute Units
12. Height & Width Property
13. Background-Image Property
14. Visibility Property

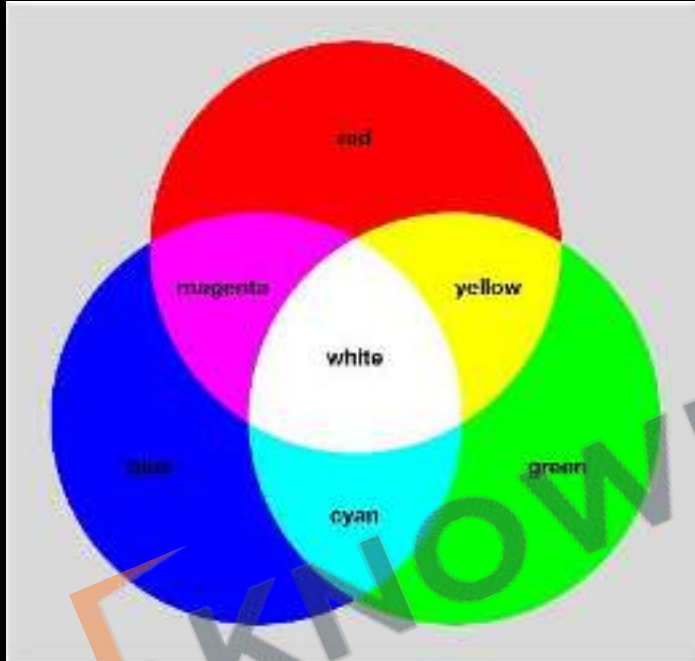
9. Background Color

```
<head>
  <title>Background Color</title>
  <style>
    #first { color: black; background-color: red; }
    #second { color: black; background-color: green; }
    button { color: black; background-color: blue; }
  </style>
</head>
<body>
  <div id="first">First</div>
  <div id="second">Second</div>
  <button>Click Me</button>
</body>
```



- **Definition:** Sets the background color of an element.
- **Syntax:** Utilized as background-color: color;
- **Visual Appeal:** Enhances the visual appeal and contrast of webpage elements.

10. Color System (Color Theory)

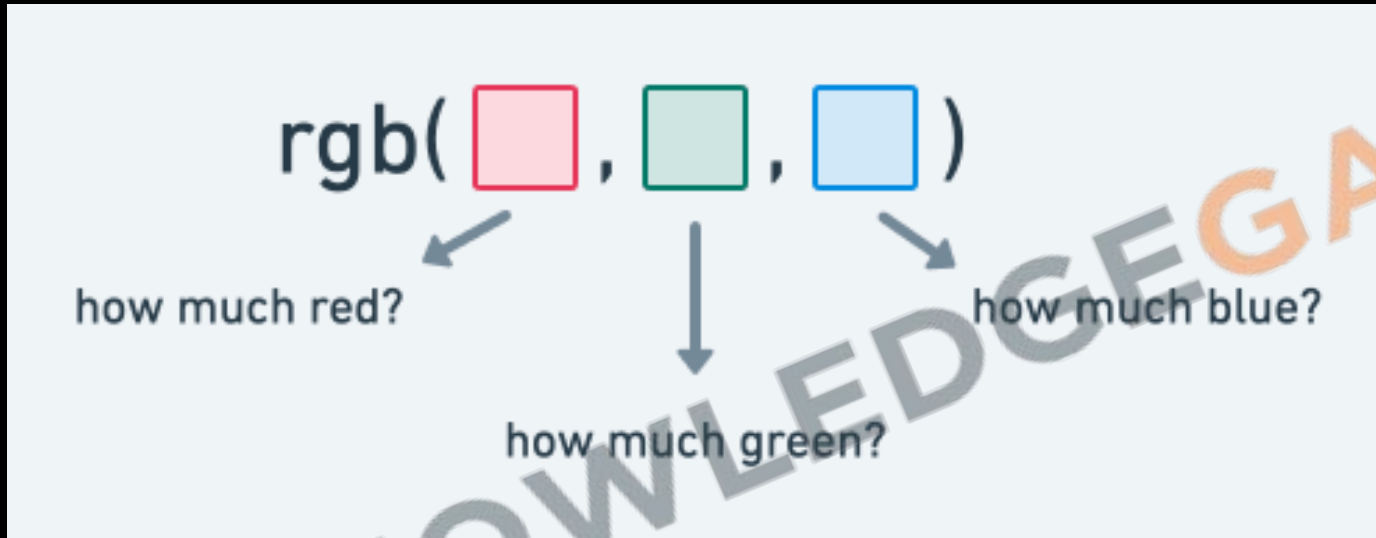


- **RGB Model:** Creates colors by mixing Red (R), Green (G), and Blue (B) light sources.
- **Additive Model:** More light means increased brightness.
- **Primary Colors:** R, G, and B are the foundational colors.
- **White & Black:** All combined yield white; absence equals black.
- **Color Depth:** Allows for millions of color variations.

10. Color System (color picker)



10. Color System (RGB Color Model)



- **Three Channels:** Consists of Red (R), Green (G), and Blue (B) channels to create a variety of colors.
- **Syntax:** Utilized as `rgb(r, g, b)` where r, g, and b are values between 0 and 255.

10. Color System (RGB Color Model)

```
<!DOCTYPE html>
<html lang="en">
<head>
  <title>RGB Color</title>
</head>
<body>
  <div style="background-color: ■ rgb(255,0,0);">First</div>
  <div style="background-color: ■ rgb(0,255,0);">Second</div>
  <div style="background-color: ■ rgb(0,0,255);">Third</div>
  <div style="background-color: ■ rgb(29, 133, 48);">Fourth</div>
</body>
</html>
```



10. Color System (HEX Color Model)

- **Hexadecimal Codes:** Represents colors using hexadecimal values, consisting of 6 digits combined from numbers and letters (A-F).
- **Syntax:** Written as #RRGGBB
- **Easy Color Matching:** Facilitates easy color matching with graphic design tools and branding colors.
- **Web Standards:** Widely supported and a common standard for defining colors in web design



10. Color System (HEX Color Model)

```
<!DOCTYPE html>
<html lang="en">
<head>
  <title>Hex Color</title>
</head>
<body>
  <div style="background-color: ■ #ff0000">First</div>
  <div style="background-color: ■ #00ff00">Second</div>
  <div style="background-color: ■ #0000ff">Third</div>
  <div style="background-color: ■ #402ae9">Fourth</div>
</body>
</html>
```



10. Color System (Alpha Channel)

- **RGBA:** RGB's extension, includes alpha for opacity control (0-1 range).
- **Transparency Control:** Facilitates the adjustment of transparency levels in colors.
- **Visual Effects:** Enables the creation of visual effects like shadows and overlays.
- **Layering:** Assists in layering elements with varying degrees of visibility.



10. Color System (Alpha Channel)

```
<h1 style="color:  rgb(255,0,0,0.1);">First</div>  
<h1 style="color:  rgb(255,0,0,0.25);">First</div>  
<h1 style="color:  rgb(255,0,0,0.5);">First</div>  
<h1 style="color:  rgb(255,0,0,0.75);">First</div>  
<h1 style="color:  rgb(255,0,0,1.0);">First</div>
```

First

First

First

First

First

11. Absolute Units

- **Definition:** Pixels (px) are fixed-size units, representing a dot on a computer screen.
- **Precision:** Allows for precise control over element dimensions.
- **Graphics & Web Design:** Commonly used in graphics and web design for setting font sizes, margins, and more.
- **Cross-Browser Consistency:** Provides consistency across different browsers.
- **High-DPI Displays:** Can vary in appearance on high-DPI (dots per inch) displays.



12. Height & Width Property

```
<head>
  <title>Height</title>
  <style>
    .box { height: 40px; width: 40px;}
    #box1 {background-color: red;}
    #box2 {background-color: blue;}
    #box3 {background-color: green;}
  </style>
</head>
<body>
  <div id="box1" class="box">Box 1</div> <br>
  <div id="box2" class="box">Box 2</div> <br>
  <div id="box3" class="box">Box 3</div>
</body>
```



- **Dimensions Control:** Used to specify the height and width of elements.
- **Unit Variability:** Can use units like pixels (px)
- **Box Model Component:** Influences padding, border, and margin.
- **Min and Max Values:** Can utilize min-height, max-height, min-width, and max-width to set restrictions on dimensions.

13. Background image Property

- **Usage:** Adds an **image** as a background to elements.
- **Syntax:** Defined using **background-image:** `url('path/to/image');`.
- **Repetition:** Control image repetition using **background-repeat**.
- **Positioning:** Adjust image position using **background-position**.
- **Size Control:** Manipulate image size using **background-size**.
- **Background-Attachment:** Sets whether the background image **scrolls** with the element or remains fixed.
- **Shorthand** (**color, image, repeat, attachment, position**)

```
<head>
  <title>Background Image</title>
  <style>
    #box1 {
      background-image: url(../images/css.png);
      height: 500px;
      width: 500px;
    }
  </style>
</head>
<body>
  <div id="box1"></div>
</body>
```



14. Visibility Property

```
<head>
  <title>Visibility</title>
  <style>
    .box { height: 40px; width: 40px;}
    #box1 {background-color: red; visibility: hidden;}
    #box2 {background-color: green; visibility: visible;}
  </style>
</head>
<body>
  <div id="box1" class="box">Box 1</div> <br>
  <div id="box2" class="box">Box 2</div> <br>
</body>
```



- **Usage:** Controls the **visibility** of elements without changing the layout.
- **Values:** Can take visible, hidden, or collapse as values.
- **Space Occupancy:** Even when hidden, the element **occupies space**.
- **Interactivity:** Hidden elements are not accessible to **user interactions**.

Level 2 Revision

Color System & Background

- 9. Background-Color Property
- 10. Color System
- 11. Absolute Units
- 12. Height & Width Property
- 13. Background-Image Property
- 14. Visibility Property



Practice Set Level 2

Color System & Background

- Create a div bar with text and background color with opacity
- Change the color of the main content
- Add background image to one div
- Use background shorthand property



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Level 3

Text Properties

15. Text-Align Property
16. Text-Decoration Property
17. Text-Transform Property
18. Line-Height Property
19. Font Properties
20. Font Family
21. Icon using fonts

15. Text-Align Property

```
<head>
  <title>Text Align</title>
  <style>
    .box {height: 100px; width: 100px;}
    #box1 {background-color: red; text-align: center;}
    #box2 {background-color: green; text-align: left;}
    #box3 {background-color: blue; text-align: right;}
  </style>
</head>
<body>
  <h3 id="box1" class="box">Box1</h3>
  <h3 id="box2" class="box">Box2</h3>
  <h3 id="box3" class="box">Box3</h3>
</body>
```



- **Usage:** Controls the horizontal alignment of text within an element.
- **Values:** Can take values like left, right, center, and justify.
- **Visual Appeal:** Enhances readability and visual appeal by organizing text neatly.

16. Text-Decoration Property

```
<head>
  <title>Text Decoration</title>
  <style>
    #box1 {text-decoration: underline;}
    #box2 {text-decoration: overline;}
    #box3 {text-decoration: line-through;}
  </style>
</head>
<body>
  <h3 id="box1" class="box">Box1</h3>
  <h3 id="box2" class="box">Box2</h3>
  <h3 id="box3" class="box">Box3</h3> <hr>
  <a href="#">link1</a> <br>
  <a href="#" style="text-decoration: none;">link2</a>
</body>
```



- **Usage:** Modifies the appearance of inline text.
- **Values:** Options include **none**, **underline**, **overline**, and **line-through**.
- **Hyperlinks:** Commonly used to remove underlines from hyperlinks for aesthetic purposes.

16. Text-Decoration Property (style)

```
<!DOCTYPE html>
<html lang="en">
<head>
  <title>Text Decoration Style</title>
  <style>
    .box {text-decoration: underline;}
    #box1 {text-decoration-style: dashed;}
    #box2 {text-decoration-style: double;}
    #box3 {text-decoration-style: solid;}
    #box4 {text-decoration-style: wavy;}
  </style>
</head>
<body>
  <h3 id="box1" class="box">Box1</h3>
  <h3 id="box2" class="box">Box2</h3>
  <h3 id="box3" class="box">Box3</h3>
  <h3 id="box4" class="box">Box4</h3>
</body>
</html>
```



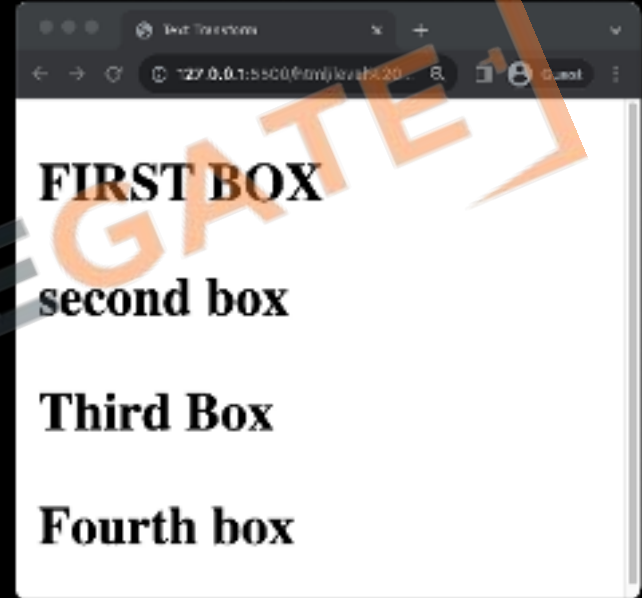
16. Text-Decoration Property (color)

```
<head>
  <title>Text Decoration Style</title>
  <style>
    .box {text-decoration: underline;}
    #box1 {text-decoration-color: red;}
    #box2 {text-decoration-color: blue;}
    #box3 {text-decoration-color: green;}
    #box4 {text-decoration-color: rgb(86, 20, 86);}
  </style>
</head>
<body>
  <h3 id="box1" class="box">Box1</h3>
  <h3 id="box2" class="box">Box2</h3>
  <h3 id="box3" class="box">Box3</h3>
  <h3 id="box4" class="box">Box4</h3>
</body>
```



17. Text-Transform Property

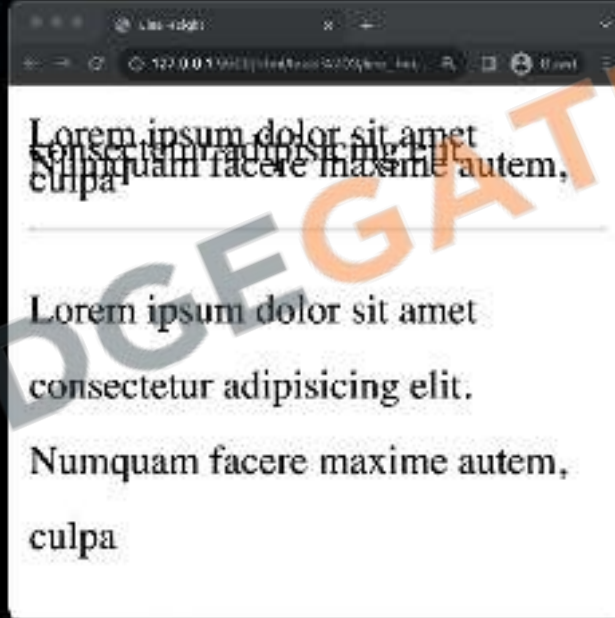
```
<head>
  <title>Text Transform</title>
  <style>
    #box1 {text-transform: uppercase;}
    #box2 {text-transform: lowercase;}
    #box3 {text-transform: capitalize;}
    #box4 {text-transform: none;}
  </style>
</head>
<body>
  <h3 id="box1" class="box">First box</h3>
  <h3 id="box2" class="box">Second box</h3>
  <h3 id="box3" class="box">Third box</h3>
  <h3 id="box4" class="box">Fourth box</h3>
</body>
```



- **Usage:** Controls the capitalization of text.
- **Common Values:** Can be uppercase, lowercase, or capitalize.
- **None Value:** none value disables text transformations.
- **Typography:** Useful for setting text style and improving typography in web design.

18. Line Height

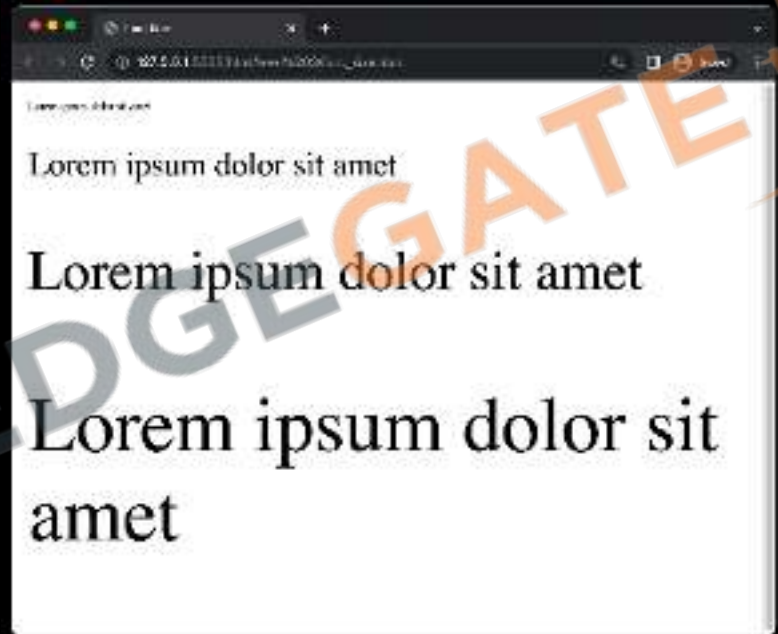
```
<head>
  <title>Line Height</title>
  <style>
    #first { line-height: 6px; }
    #second { line-height: 30px; }
  </style>
</head>
<body>
  <p id="first">Lorem ipsum dolor sit amet
consectetur adipisicing elit. Numquam facere
maxime autem, culpa</p> <hr>
  <p id="second">Lorem ipsum dolor sit amet
consectetur adipisicing elit. Numquam facere
maxime autem, culpa</p>
</body>
```



- **Usage:** Adjusts the amount of space above and below inline elements.
- **Readability:** Enhances text readability by preventing overcrowding.
- **Vertical Spacing:** Useful for controlling vertical spacing between lines of text.

19. Font Property (font-size)

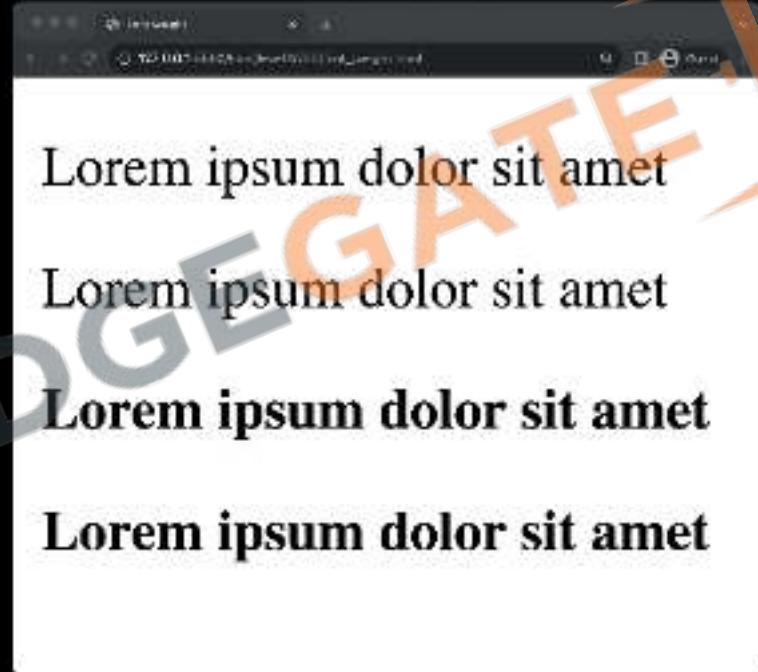
```
<head>
  <title>Font Size</title>
  <style>
    #first {font-size: 5px;}
    #second {font-size: 15px;}
    #third {font-size: 25px;}
    #fourth {font-size: 35px;}
  </style>
</head>
<body>
  <p id="first">Lorem ipsum dolor sit amet</p>
  <p id="second">Lorem ipsum dolor sit amet</p>
  <p id="third">Lorem ipsum dolor sit amet</p>
  <p id="fourth">Lorem ipsum dolor sit amet</p>
</body>
```



- **Usage:** Sets the **size of the font** in web content.
- **Responsiveness:** Helps in creating **responsive designs** adaptable to various screen sizes.
- **Readability:** Crucial for ensuring the **readability** of text on websites.

19. Font Property (font-weight)

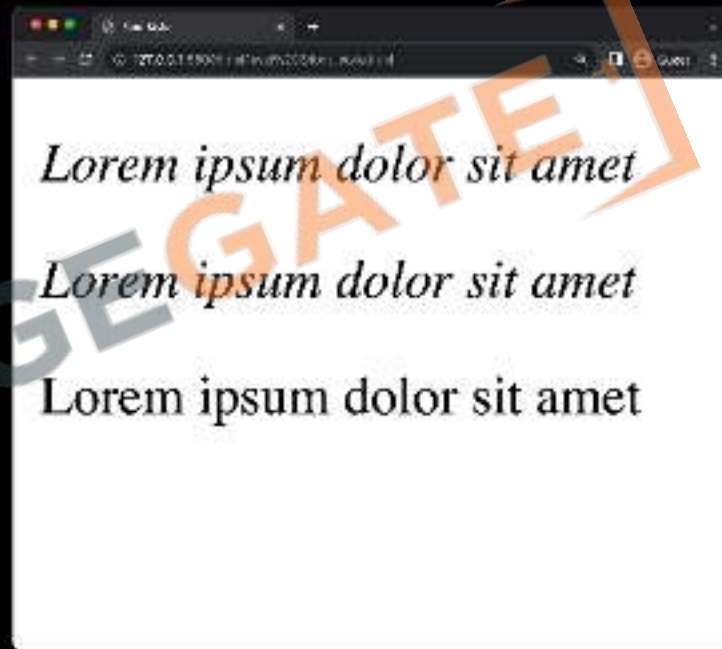
```
<head>
  <title>Font Weight</title>
  <style>
    #first {font-weight: 100;}
    #second {font-weight: 400;}
    #third {font-weight: 600;}
    #fourth {font-weight: 900;}
  </style>
</head>
<body>
  <p id="first">Lorem ipsum dolor sit amet</p>
  <p id="second">Lorem ipsum dolor sit amet</p>
  <p id="third">Lorem ipsum dolor sit amet</p>
  <p id="fourth">Lorem ipsum dolor sit amet</p>
</body>
```



- **Usage:** Defines the **thickness** of characters in a font.
- **Values:** Can take values like **normal**, **bold**, **bolder**, or numeric values (**100 to 900**).
- **Text Emphasis:** Utilized to emphasize text or **create contrast** between text elements.

19. Font Property (font-style)

```
<head>
  <title>Font Style</title>
  <style>
    #first {font-style: italic;}
    #second {font-style: oblique;}
    #third {font-style: normal;}
  </style>
</head>
<body>
  <p id="first">Lorem ipsum dolor sit amet</p>
  <p id="second">Lorem ipsum dolor sit amet</p>
  <p id="third">Lorem ipsum dolor sit amet</p>
</body>
```



- **Usage:** Controls the **style** of the font, mainly affecting its inclination.
- **Values:** Common values are **normal**, **italic**, and **oblique**.
- **Text Formatting:** Useful for **highlighting** or distinguishing certain text segments.

20. Font Family

- **Usage:** Defines which font should be used for text within an element.
- **Specific Fonts:** Common choices include Arial, Segoe UI, Times New Roman, and others.
- **Fallback Mechanism:** Incorporate a fallback font family in case the primary font is unavailable; helps in maintaining the site aesthetics.
- **Web Safe Fonts:** Employ web-safe fonts to ensure consistency across different browsers and operating systems.
- **Generic Family:** Always end the font family list with a generic family like serif or sans-serif as a last resort option.

Arial Narrow
Book Antiqua
Cambria
Century Gothic
Consolas
COPPERPLATE
Georgia
Impact
Lucida Sans Unicode
Papyrus
Script MT Bold
Tahoma
Times New Roman
Verdana

20. Font Family

```
<head>
  <title>Font Family</title>
  <link rel="preconnect" href="https://fonts.googleapis.com">
  <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
  <link href="https://fonts.googleapis.com/css2?family=Oswald:wght@200&
display=swap" rel="stylesheet">
  <style>
    #first {font-family: Arial, Helvetica, sans-serif;}
    #second {font-family: 'Segoe UI', Tahoma, Geneva, Verdana,
    sans-serif;}
    #third {font-family: 'Oswald', sans-serif;}
  </style>
</head>
<body>
  <p id="first">Lorem ipsum dolor sit amet</p>
  <p id="second">Lorem ipsum dolor sit amet</p>
  <p id="third">Lorem ipsum dolor sit amet</p>
</body>
```

Lorem ipsum dolor sit amet

Lorem ipsum dolor sit amet

Lorem ipsum dolor sit amet

21. Icons using Fonts

```
<head>
  <title>Icons</title>
  <script src="https://kit.fontawesome.com/43290fa92d.
  js" crossorigin="anonymous"></script>
</head>
<body>
  <i class="fa-solid fa-house"></i>
  <i class="fa-brands fa-facebook"></i>
  <i class="fa-brands fa-tiktok"></i>
  <i class="fa-solid fa-poo" style="font-size: 40px;"></i>
</body>
```



Using <https://fontawesome.com>

Level 3 Revision

Text Properties

- 15. Text-Align Property
- 16. Text-Decoration Property
- 17. Text-Transform Property
- 18. Line-Height Property
- 19. Font Properties
- 20. Font Family
- 21. Icon using fonts



Practice Set Level 3

Text Properties

- Create an **Heading** at the centre and make capitalized
- Use Font family for the whole page to **TimeNewRoman**
- Create one div inside another div. Set id and text **outer** to outer div, set id and text **inner** to inner div. Set outer div text size to 25px. Set inner to 10px.
- Use icons from **fontawesome.com** and use icons of **LinkedIn** and **GitHub**



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Level 4

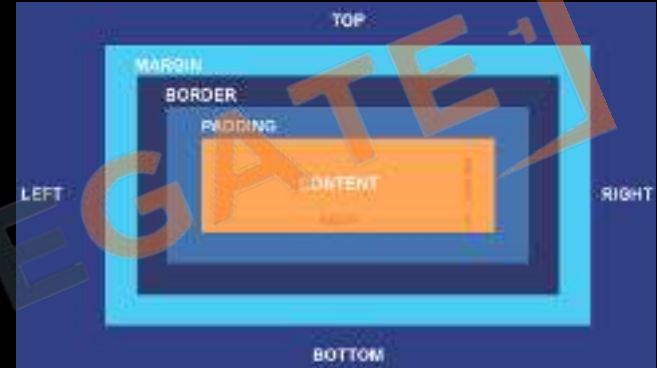
Box Model

- 22. What is Box Model
- 23. Padding Property
- 24. Margin Property
- 25. Border Property

22. What is Box Model

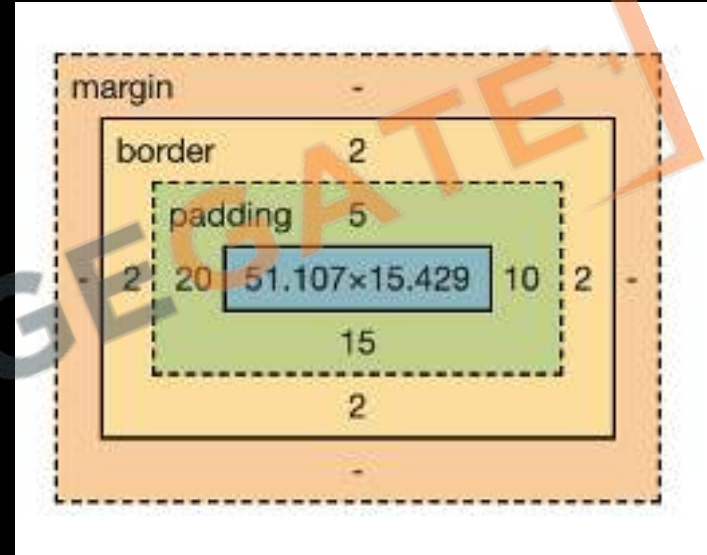
मिठाई भण्डार पे चलो

- **Core Concept:** Central concept in CSS that outlines the design and layout of elements on the web page.
- **Components:** Consists of four main components - **margin**, **border**, **padding**, and **content**.
- **Margin:** The space outside the border, separating the element from others.
- **Border:** The outline that encapsulates the padding and content.
- **Padding:** The space between the border and the actual content, providing a buffer.
- **Content:** The innermost layer where text, images, or other media are housed.



23. Padding Property

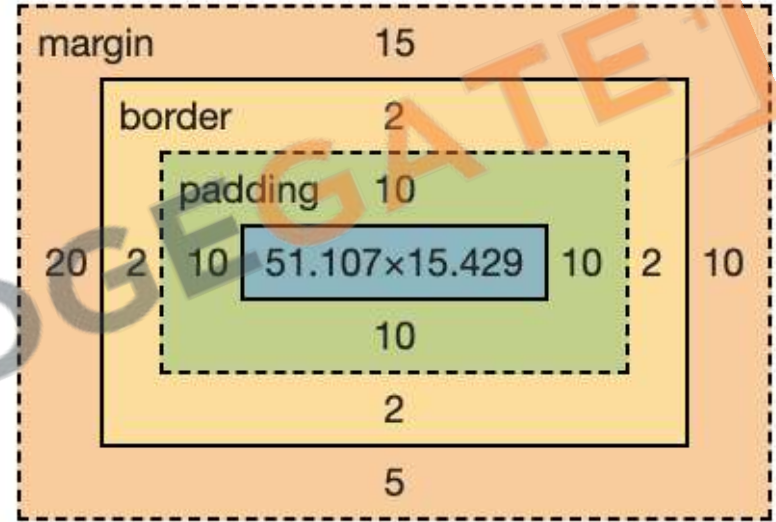
```
<head>
  <title>Padding</title>
  <style>
    * { margin: 0; padding: 0; }
    #button1 {
      padding: 5px 10px 15px 20px;
      background-color: aquamarine;
    }
  </style>
</head>
<body>
  <button id="button1">Click Me</button>
</body>
```



- **Usage:** Defines the space between the content of an element and its border.
- **Individual Sides:** Allows setting padding for individual sides using `padding-top`, `padding-right`, `padding-bottom`, and `padding-left`.
- **Shorthand:** Can use shorthand property `padding` to set all sides at once, e.g., `padding: 10px 20px 10px 20px`.

24. Margin Property

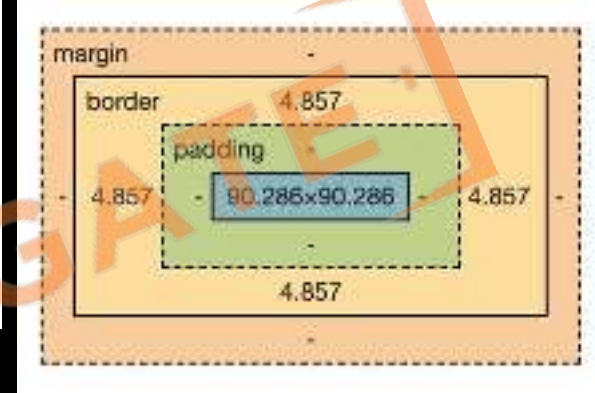
```
<head>
  <title>margin</title>
  <style>
    * { margin: 0; padding: 0; }
    #button1 {
      margin: 15px 10px 5px 20px;
      padding: 10px;
      background-color: aquamarine;
    }
  </style>
</head>
<body>
  <button id="button1">Click Me</button>
</body>
```



- **Functionality:** Sets the space around elements, separating them from others.
- **Individual Sides:** Customizable for **top**, **right**, **bottom**, and **left** sides.
- **Shorthand:** Allows quick setup, e.g., **margin: 10px 20px**. (clockwise)
- **Auto Value:** Can be used for central alignment with **auto** value.

24. Border Property

```
<head>
  <title>Border</title>
  <style>
    * { margin: 0; padding: 0; }
    #button1 {
      height: 100px; width: 100px;
      border: 5px dashed black;
      background-color: aquamarine;
    }
  </style>
</head>
<body>
  <button id="button1">Click Me</button>
</body>
```



- **Usage:** Creates an outline around HTML elements.
- **Components:** Defined by width, style, and color attributes.
- **Styles:** Includes options like solid, dashed, and dotted.
- **Shorthand:** Can set attributes at once, e.g., border: 2px solid black.

24. Border Property (border radius)

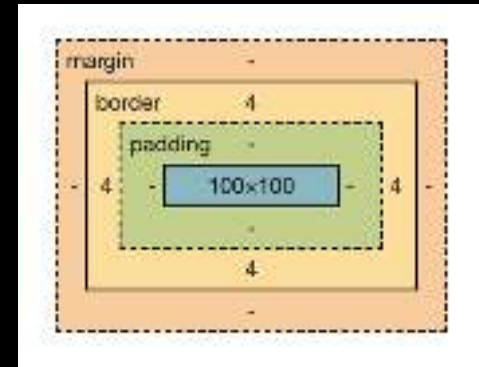
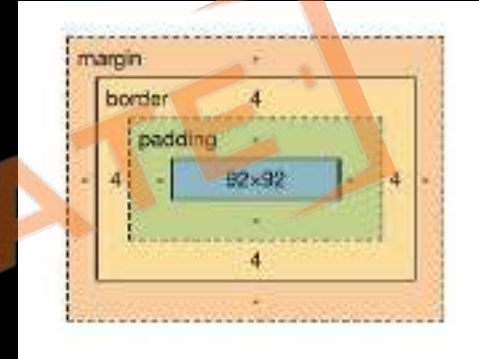
```
<head>
  <title>Border Radius</title>
  <style>
    * { margin: 0; padding: 0; }
    #button1 {
      height: 100px; width: 100px;
      border: 5px solid black;
      border-radius: 30px;
      background-color: aquamarine;
    }
  </style>
</head>
<body>
  <button id="button1">Click Me</button>
</body>
```



- **Usage:** Used to create rounded corners for elements.
- **Individual Corners:** Allows setting different radii for each corner.
- **Shorthand:** e.g., border-radius: 10px 20px.

24. Border Property (box sizing)

```
<head>
  <title>Box Sizing</title>
  <style>
    * { margin: 0; padding: 0; }
    button {
      height: 100px; width: 100px;
      border: 4px solid black;
      background-color: aquamarine;
    }
    #button1 { box-sizing: border-box; }
    #button2 { box-sizing: content-box; }
  </style>
</head>
<body>
  <button id="button1">Click Me</button>
  <button id="button2">Click Me</button>
</body>
```



Level 4 Revision

Box Model

- 22. What is Box Model
- 23. Padding Property
- 24. Margin Property
- 25. Border Property



Practice Set Level 4

Box Model

- Create a div with height and width 200px and observe different values in the box model by inspecting.
- Create a button and give 10px padding to top and bottom and 15px on the sides.
- Add 100px margin to the button on all sides.
- Add dotted border with color red and so much that the button becomes a circle.
- Create two boxes with different box-sizing values and observe changes in box model.



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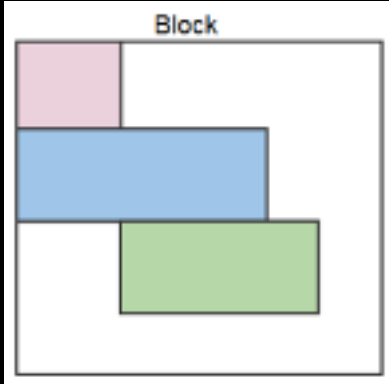
[Sanchit Socket](#)

Level 5

Display and Position

- 26. Display Property
- 27. Responsive Websites
- 28. Relative Units
- 29. Position Property
- 30. Semantic Tags

26. Display Property (Block / Inline Elements)



Block Elements

- **New Line:** Start on a new line.
- **Full Width:** Take up all horizontal space.
- **Styling:** Can have margins and padding.
- **Size:** Width and height can be set.
- **Examples:** `<div>`, `<p>`, `<h1>`, `<ul>`, `<li>`.

Inline Elements

- **Flow:** Stay in line with text.
- **Width:** Just as wide as the content.
- **No Break:** No new line between elements.
- **Limited Styling:** Can't set size easily.
- **Examples:** `<span>`, `<a>`, `<strong>`, `<em>`.

26. Display Property (Block)

```
.box {  
  height: 100px;  
  width: 100px;  
  background-color: blueviolet;  
  margin: 10px;  
  text-align: center;  
  border: 5px solid black;  
  border-radius: 10px;  
  display: block;  
}
```



```
<head>  
  <title>Display Block</title>  
  <link rel="stylesheet" href="../../css/level 5/display.css">  
</head>  
<body>  
  <div id="parent">  
    <div id="div1" class="box">box1</div>  
    <div id="div2" class="box">box2</div>  
    <div id="div3" class="box">box3</div>  
  </div>  
</body>
```

26. Display Property (Inline)

```
.box {  
  height: 100px;  
  width: 100px;  
  background-color: blueviolet;  
  margin: 10px;  
  text-align: center;  
  border: 5px solid black;  
  border-radius: 10px;  
  display: inline;  
}
```

```
<!DOCTYPE html>  
<html lang="en">  
<head>  
  <title>Display Inline</title>  
  <link rel="stylesheet" href="../../css/level 5/display inline.css">  
</head>  
<body>  
  <div id="parent">  
    <div id="div1" class="box">box1</div>  
    <div id="div2" class="box">box2</div>  
    <div id="div3" class="box">box3</div>  
  </div>  
</body>  
</html>
```

box1

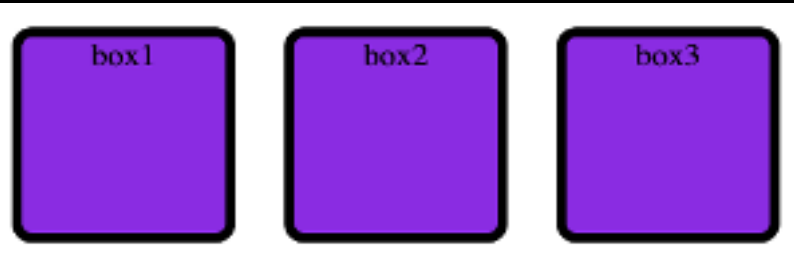
box2

box3

26. Display Property (Inline-Block)

```
.box {  
  height: 100px;  
  width: 100px;  
  background-color: blueviolet;  
  margin: 10px;  
  text-align: center;  
  border: 5px solid black;  
  border-radius: 10px;  
  display: inline-block;  
}
```

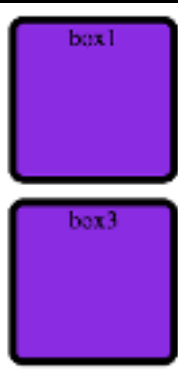
```
<head>  
  <title>Display Inline Block</title>  
  <link rel="stylesheet" href="../../css/  
    level 5/display_inline_block.css">  
</head>  
<body>  
  <div id="parent">  
    <div id="div1" class="box">box1</div>  
    <div id="div2" class="box">box2</div>  
    <div id="div3" class="box">box3</div>  
  </div>  
</body>
```



26. Display Property (None)

```
.box {  
  height: 100px;  
  width: 100px;  
  background-color: blueviolet;  
  margin: 10px;  
  text-align: center;  
  border: 5px solid black;  
  border-radius: 10px;  
}  
  
#div2 {  
  display: none;  
}
```

```
<head>  
  <title>Display None</title>  
  <link rel="stylesheet" href="../../css/  
    level 5/display_none.css">  
</head>  
<body>  
  <div id="parent">  
    <div id="div1" class="box">box1</div>  
    <div id="div2" class="box">box2</div>  
    <div id="div3" class="box">box3</div>  
  </div>  
</body>
```



27. Responsive Website



27. Responsive Website



1. Adapts layout for different screen sizes
2. Flexible layouts
3. Optimizes images and assets
4. Enhances user experience on mobile and desktop

28. Relative Units



CSS Units Cheat Sheet

px

Absolute pixel value

%

A percentage of the parent element.
100% is the width of the parent element

em

Relative to the font size of the element

vh

Relative to 1% of the viewport's height

rem

Relative to the font size of
the root element

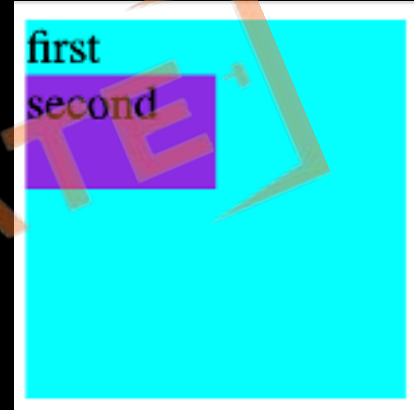
vw

Relative to 1% of the viewport's width

28. Relative Units (Percentage)

```
#first {  
  height: 200px;  
  width: 200px;  
  background-color: aqua;  
  font-size: 25px;  
}  
  
#second {  
  background-color: blueviolet;  
  width: 50%;  
  height: 30%;  
}
```

```
<body>  
  <div id="first">  
    first  
    <div id="second">  
      second  
    </div>  
  </div>  
</body>
```



- **Relative Sizing:** Facilitates dynamic sizing **relative to parents**.
- **Adaptability:** Ensures **responsiveness** across various screens.
- **Dimensions:** Quickly set width and height as a percentage.

28. Relative Units (EM)

```
body {  
  font-size: 100px;  
}  
  
#first {  
  height: 200px;  
  width: 200px;  
  background-color: aqua;  
  font-size: 25px;  
}
```

```
#second {  
  background-color: blueviolet;  
  width: 70%;  
  height: 50%;  
  font-size: 2em;  
}
```

```
<body>  
  Body  
  <div id="first">  
    first  
    <div id="second">  
      second  
    </div>  
  </div>  
</body>
```



- **Relative Unit:** Sized **relative to the parent** element's font size.
- **Scalability:** Facilitates easy **scaling** of elements for responsive design.
- **Font Sizing:** Commonly used for setting font sizes adaptively.

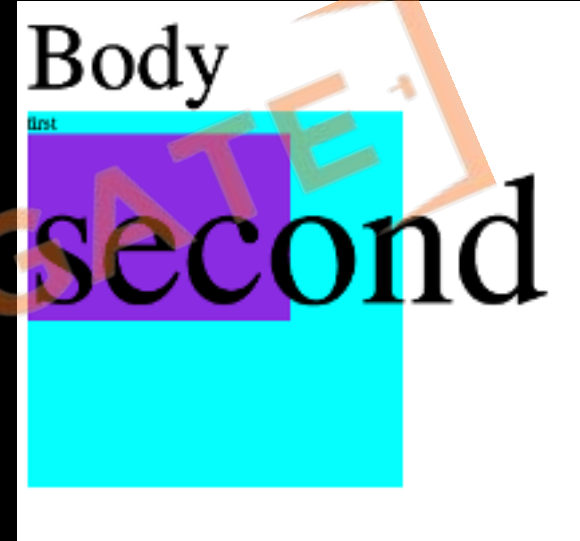
28. Relative Units (REM)

```
* {
  font-size: 50px;
}

#first {
  height: 200px;
  width: 200px;
  background-color: aqua;
  font-size: 10px;
}

#second {
  background-color: blueviolet;
  width: 70%;
  height: 50%;
  font-size: 2rem;
}
```

```
<body>
  Body
  <div id="first">
    first
    <div id="second">
      second
    </div>
  </div>
</body>
```



- **Relative Sizing:** Facilitates dynamic sizing **relative to root element**.
- **Adaptability:** Ensures **responsiveness** across various screens.
- **Dimensions:** Quickly set width and height as a percentage.

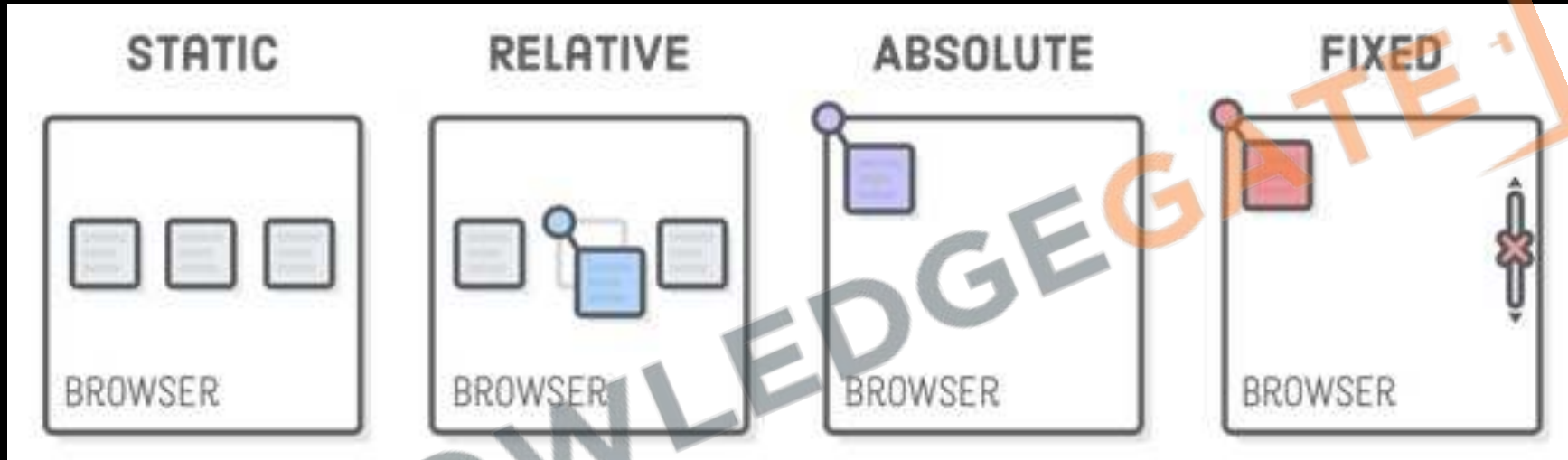
28. Relative Units (vw/vh)

```
<head>
  <title>Unit VH and VW</title>
  <style>
    #first {
      height: 50vh;
      width: 90vw;
      background-color: red;
    }
  </style>
</head>
<body>
  <div id="first"></div>
</body>
```



- **Viewport Relative Units:** Units based on **viewport's width (vw)** or **height (vh)** for responsive design.
- **Responsive Layouts:** Essential for creating adaptive layouts; e.g., **height: 100vh** for **full-screen** sections.
- **Element Sizing:** Useful for defining heights and widths that **scale with the viewport**.

29. Position Property

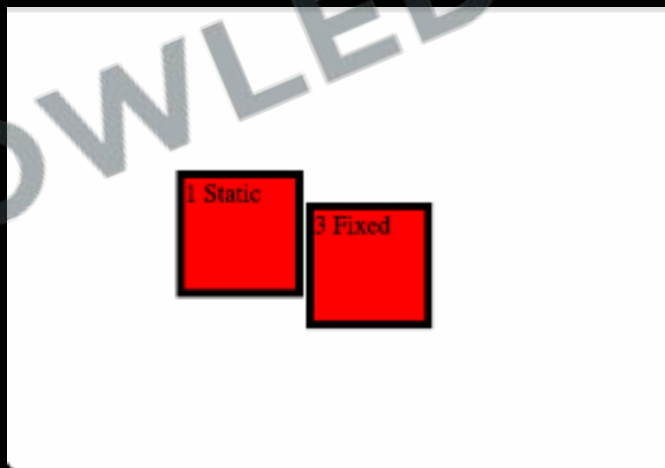


- **Static (default)** : Elements follow the normal document flow. (top, right, bottom, left, z-index would not work)
- **Relative**: Element's position adjusted from its normal position.
- **Absolute**: Positions element relative to the nearest positioned ancestor.
- **Fixed**: Element positioned relative to the viewport, does not move on scroll.

29. Position Property

```
div {  
  height: 70px;  
  width: 70px;  
  background-color: red;  
  border: 5px solid black;  
  margin: 20px;  
}  
#div1 {  
  position: static;  
}  
#div2 {  
  position: relative;  
  top: 20px;  
  left: 90px;  
}  
#div3 {  
  position: fixed;  
  top: 20px;  
  left: 90px;  
}  
#div4 {  
  position: absolute;  
  top: 200px;  
  left: 200px;  
}
```

```
<head>  
  <title>Position</title>  
  <link rel="stylesheet" href="../../css/level 5/positions.css">  
</head>  
<body>  
  <div id="div1">1 Static</div>  
  <div id="div2">2 Relative</div>  
  <div id="div3">3 Fixed</div>  
  <div id="div4">4 Absolute</div>  
</body>
```

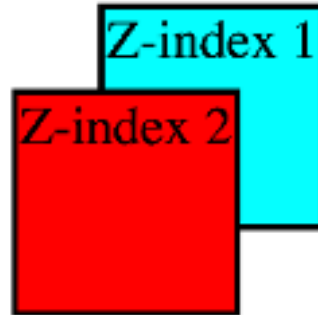


29. Position Property (z index)

```
.container {  
  position: relative;  
}  
.box1, .box2 {  
  position: absolute;  
  border: 3px solid black;  
  width: 100px;  
  height: 100px;  
  text-align: center;  
  font-size: 25px;  
}  
.box1 {  
  background-color: red;  
  left: 20px;  
  top: 60px;  
  z-index: 2;  
}  
.box2 {  
  background-color: aqua;  
  left: 60px;  
  top: 20px;  
  z-index: 1;  
}
```

- **Stacking Order:** Determines the **stacking order** of elements along the Z-axis.
- **Position Context:** Only applies to elements with position set to relative, absolute, fixed, or sticky.
- **Integer Values:** Accepts **integer values**, including negative numbers.
- **Higher Values:** An element with a **higher z-index value appears above others**.

```
<head>  
  <title>Z-Index</title>  
  <link rel="stylesheet" href="../../css/level 5/z-index.css">  
</head>  
<body>  
  <div class="container">  
    <div class="box1">Z-index 2</div>  
    <div class="box2">Z-index 1</div>  
  </div>  
</body>
```



30. Semantic Tags



30. Semantic Tags

Semantic Tags

- **Meaningful**: Describe content.
- **SEO**: Good for search engines.
- **Accessibility**: Useful for screen readers.
- Examples: `<header>`, `<footer>`, `<article>`, `<section>`, `<nav>`.

Non-Semantic Tags

- **Generic**: No specific meaning.
- **For Styling**: Used for layout.
- **No SEO**: Not SEO-friendly.
- Examples: `<div>`, `<span>`, `<i>`, `<b>`.

Level 5 Revision

Display and Position

- 26. Display Property
- 27. Responsive Websites
- 28. Relative Units
- 29. Position Property
- 30. Semantic Tags



Practice Set Level 5

Display and Position

- Create a webpage with header, footer, and a content area.
 - Header
 - Create a nav bar with links.
 - Main
 - Create a div with width and height, Background green and border radius 50%
 - Create Three divs with container height and width 100px. Display inline block.
 - Set the correct position property for the single div element to ensure it remains at the right side of the page and does not shift when scrolling.
 - Use z-index to place the div on top of another div.
 - Footer
 - Add text in footer.



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Level 6

Flex Box, Grid and Media Queries

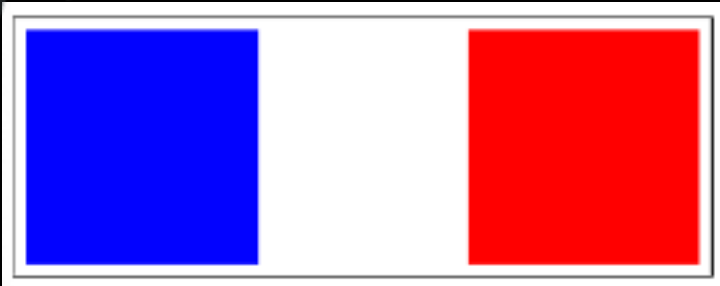
- 31. Float Property
- 32. What is Flexbox?
- 33. Flex Model
- 34. Flexbox Direction
- 35. Properties: Flexbox container
- 36. Properties: Flex Items
- 37. Grid Layout
- 38. Media Queries

31. Float Property

```
.container {  
  height: 110px;  
  width: 300px;  
  border: 1px solid black;  
}  
  
.box {  
  width: 100px;  
  height: 100px;  
  margin: 5px;  
}  
  
.box1 {  
  background-color: red;  
  float: right;  
}  
  
.box2 {  
  background-color: blue;  
  float: left;  
}
```

- **Element Alignment:** Allows elements to be aligned to the left or right within their containing element.
- **Values:** Can take values like "left", "right", or "none" to determine the floating direction.
- **Old Layout Technique:** Less commonly used with the advent of Flexbox.

```
<html>  
  <head>  
    <title>Float Property</title>  
    <link rel="stylesheet" href="css/level 6/float.css">  
  </head>  
  <body>  
    <div class="container">  
      <div class="box box1"></div>  
      <div class="box box2"></div>  
    </div>  
  </body>  
</html>
```

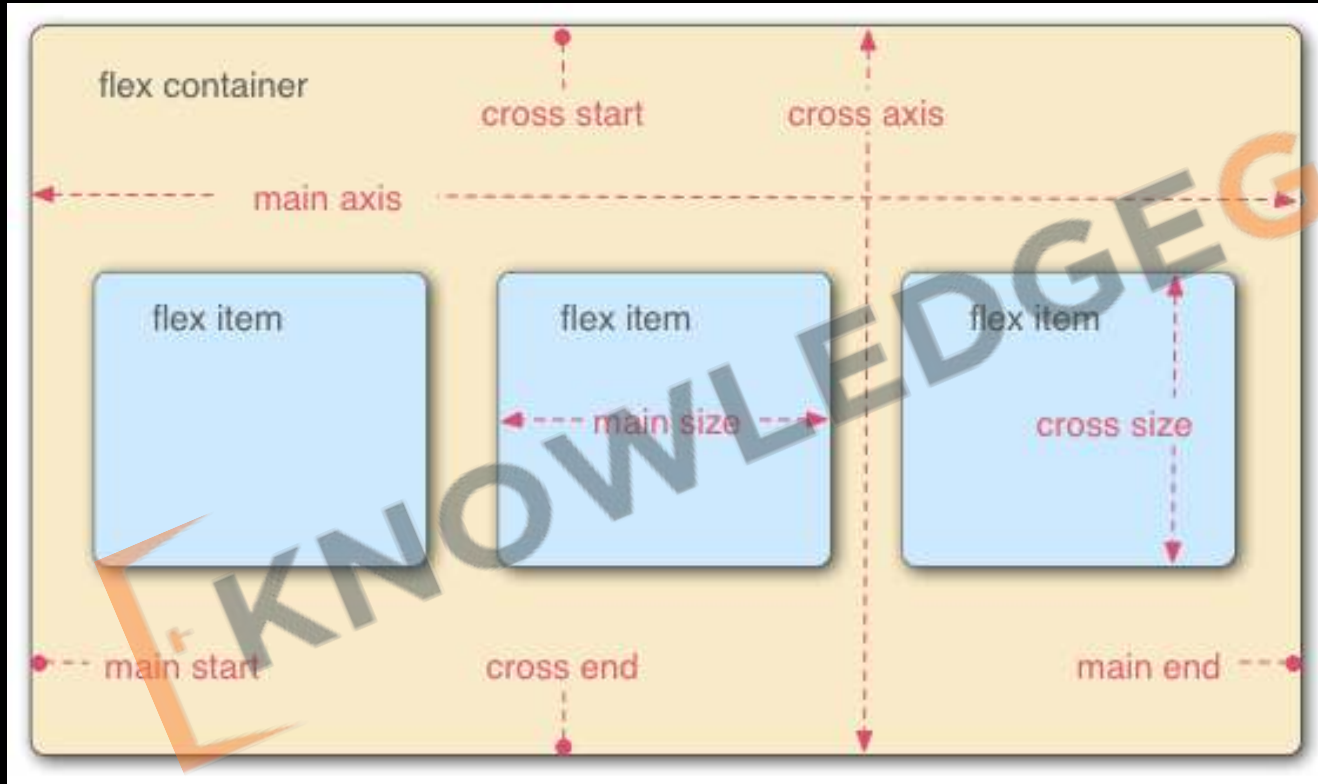


32. What is Flexbox?

Flexbox is a **one-dimensional layout** method for **arranging items in rows or columns**. Items *flex* (expand) to fill additional space or shrink to fit into smaller spaces.



33. Flex Model



display: flex

34. Flexbox Direction

- **Property Name:** flex-direction is the property used to define the direction in a flex container.
- **Row Layout:** row value aligns the flex items **horizontally**, in a left-to-right fashion.
- **Column Layout:** column value stacks the flex items **vertically**, from top to bottom.
- **Reverse Direction:** Adding -reverse to row or column (as in row-reverse or column-reverse) reverses the order of the items.



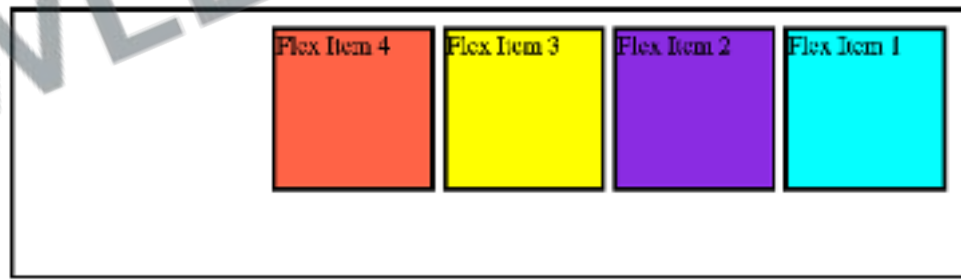
34. Flexbox Direction

```
{
  margin: 0;
  padding: 0;
}
.box {
  height: 100px;
  width: 100px;
  border: 3px solid black;
  margin-right: 5px;
}
#heading {margin-left: 200px;}
#container {
  height: 150px;
  width: 600px;
  padding: 10px;
  margin: 20px;
  border: 3px solid black;

  display: flex;
  flex-direction: row-reverse;
}
#box1 { background-color: aqua;}
#box2 { background-color: blueviolet;}
#box3 { background-color: yellow;}
#box4 { background-color: tomato;}
```

```
<body>
  <h1 id="heading">Flex Container</h1>
  <div id="container">
    <div id="box1" class="box">Flex Item 1</div>
    <div id="box2" class="box">Flex Item 2</div>
    <div id="box3" class="box">Flex Item 3</div>
    <div id="box4" class="box">Flex Item 4</div>
  </div>
</body>
```

Flex Container



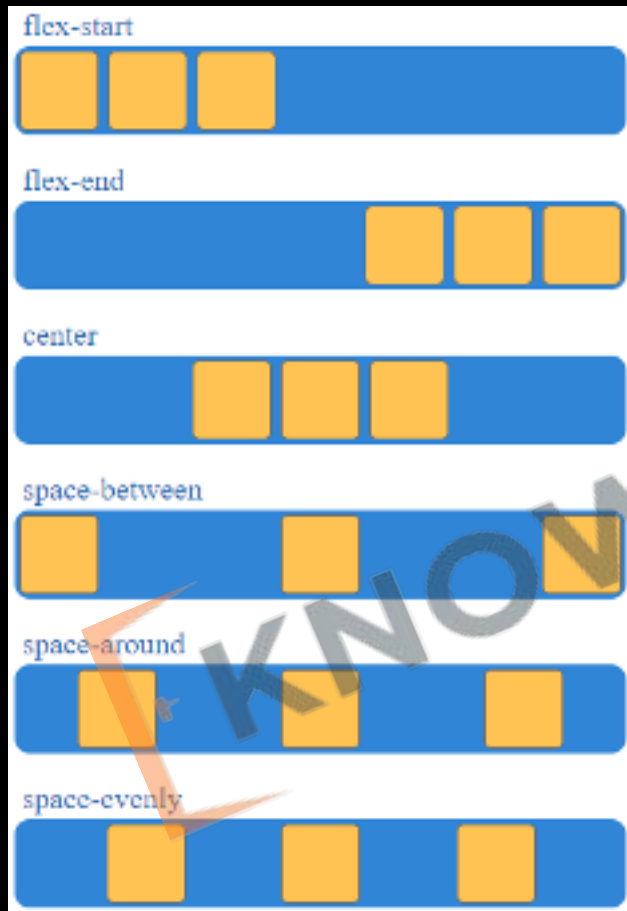
34. Flexbox Direction

```
.box {  
  height: 100px;  
  width: 100px;  
  border: 3px solid black;  
  margin-right: 5px;  
}  
#heading {margin-left: 50px;}  
#container {  
  height: 150px;  
  width: 200px;  
  padding: 10px;  
  margin: 20px;  
  border: 3px solid black;  
  
  display: flex;  
  flex-direction: row-reverse;  
}
```

Flex Container



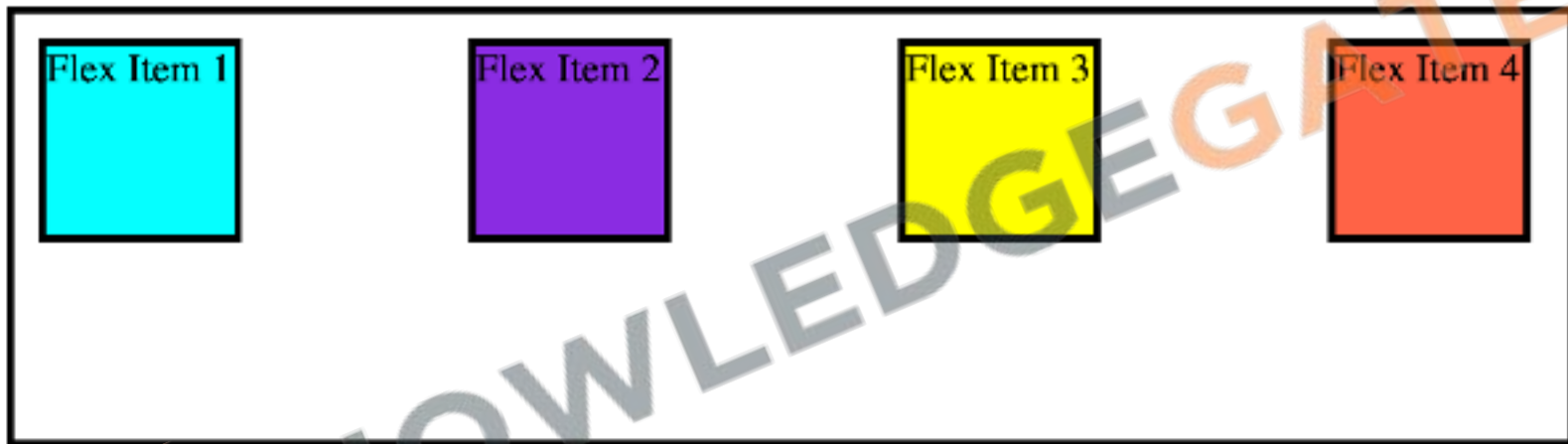
35. Properties: Flexbox container (Justify Content)



- **Alignment:** Aligns flex items along the main axis.
- **flex-start:** Items align to the start of the flex container.
- **flex-end:** Items align to the end of the flex container.
- **Center:** Items are centered within the flex container.
- **space-between/space-around/space-evenly:** Distributes space between items evenly.

35. Properties: Flexbox container (Justify Content)

Flexbox Container

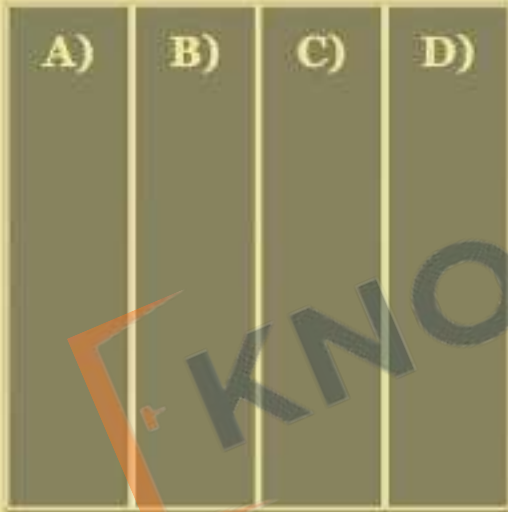


```
display: flex;  
justify-content: space-between;
```

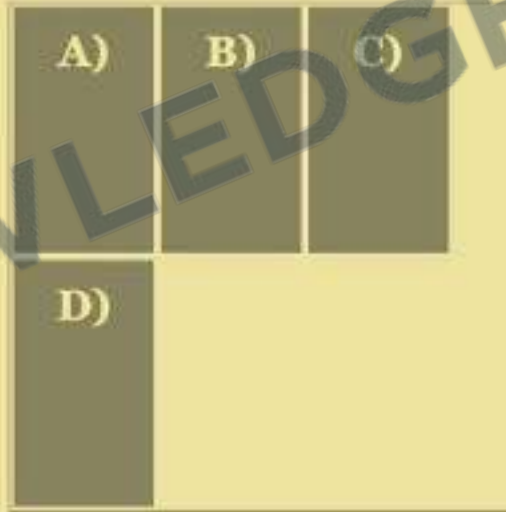
35. Properties: Flexbox container (Flex Wrap)

flex-wrap:

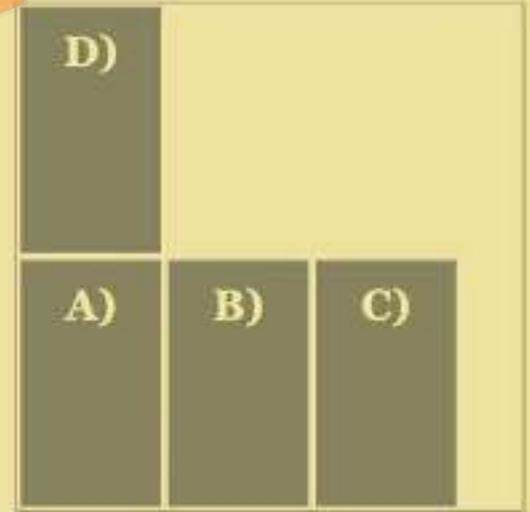
nowrap (default)



wrap



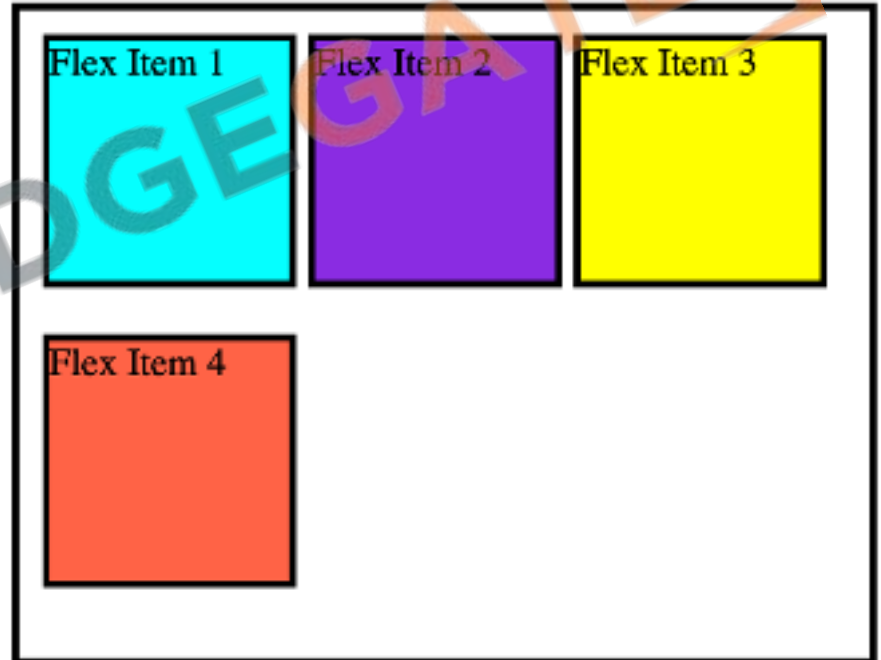
wrap-reverse



35. Properties: Flexbox container (Flex Wrap)

```
#container {  
  height: 250px;  
  width: 335px;  
  padding: 10px;  
  margin: 20px;  
  border: 3px solid black;  
  
  display: flex;  
  flex-wrap: wrap;  
}
```

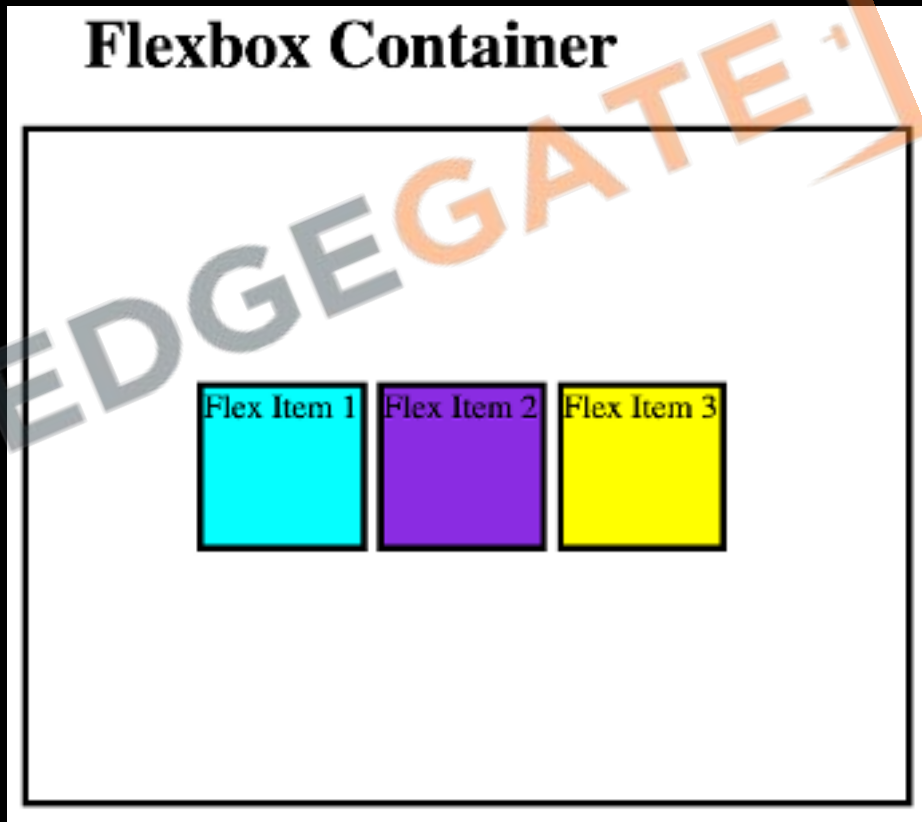
Flex Container



35. Properties: Flexbox container (Align Items)

This property is used to align the flex container's items along the cross-axis, which is perpendicular to the main axis.

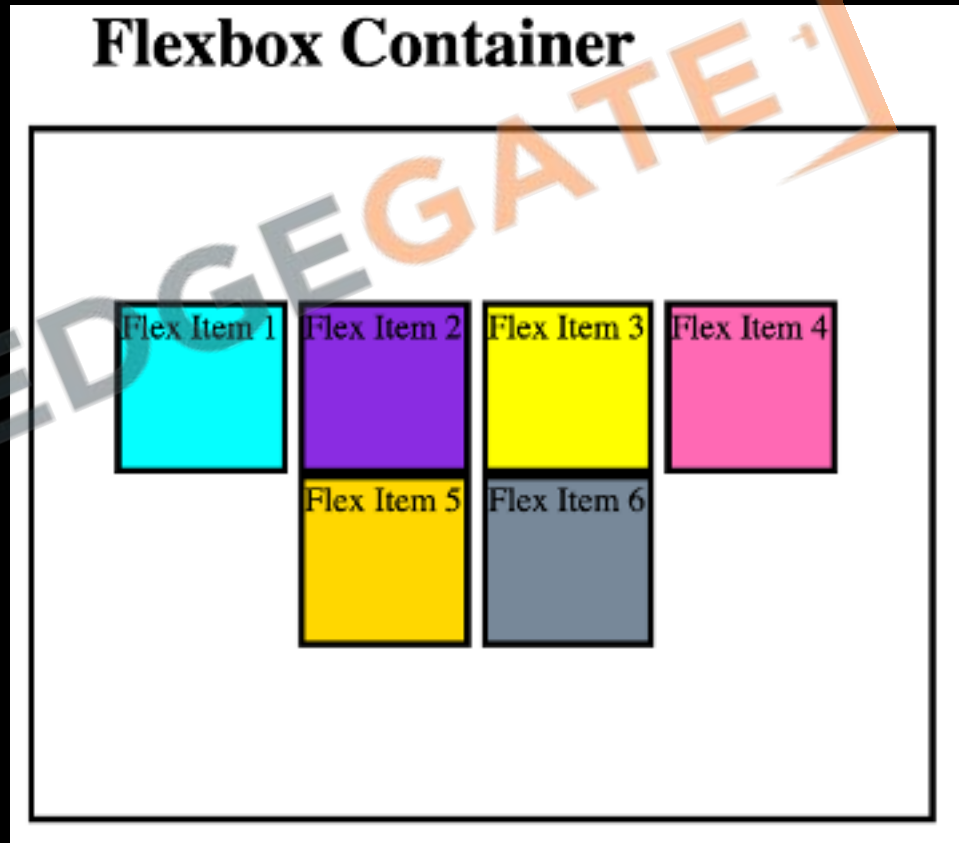
```
display: flex;  
flex-direction: row;  
justify-content: center;  
align-items: center;
```



35. Properties: Flexbox container (Align Content)

It is utilized to adjust the spacing between flex lines within a flex container, particularly when there is extra space along the cross-axis.

```
display: flex;  
flex-direction: row;  
justify-content: center;  
flex-wrap: wrap;  
align-content: center;
```

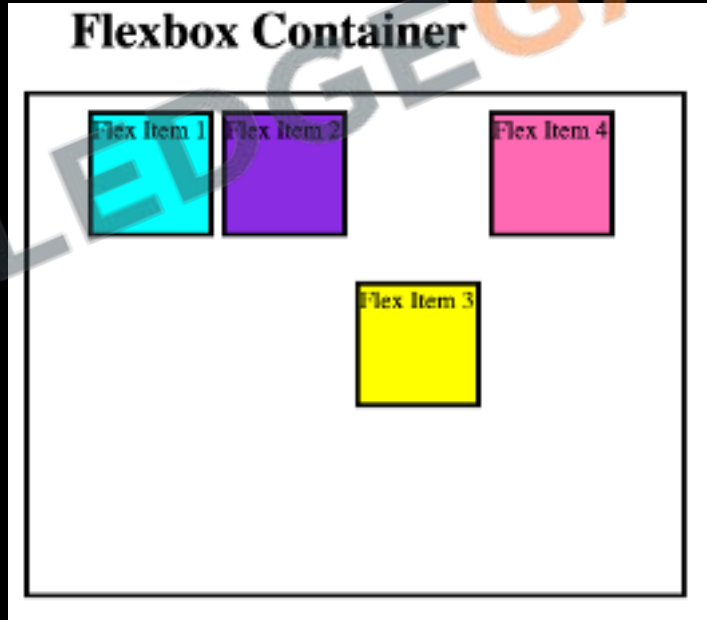


36. Properties: Flex Items (Align Self)

```
#container {  
  height: 300px;  
  width: 400px;  
  padding: 10px;  
  margin: 20px;  
  border: 3px solid black;  
  
  display: flex;  
  flex-direction: row;  
  justify-content: center;  
  flex-wrap: wrap;  
  align-items: start;  
}
```

```
#box1 { background-color: aqua;}  
#box2 { background-color: blueviolet;}  
#box4 { background-color: hotpink;}  
#box3 {  
  background-color: yellow;  
  align-self: center;  
}
```

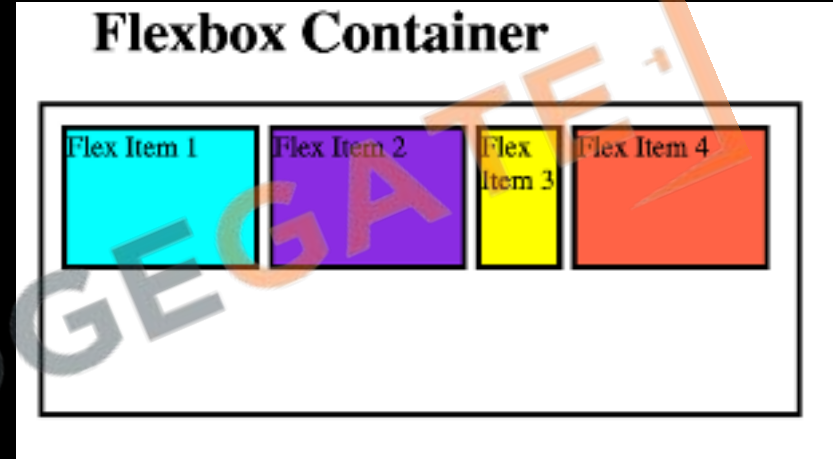
Allows individual flex items to **override the container's align-items** property, aligning them differently along the cross-axis.



36. Properties: Flex Items (Flex Shrink)

```
#box3 {  
  background-color: yellow;  
  flex-shrink: 4;  
}
```

The "flex-shrink" property in CSS determines how much a **flex item** will shrink relative to other items in the flex container if there is insufficient space.



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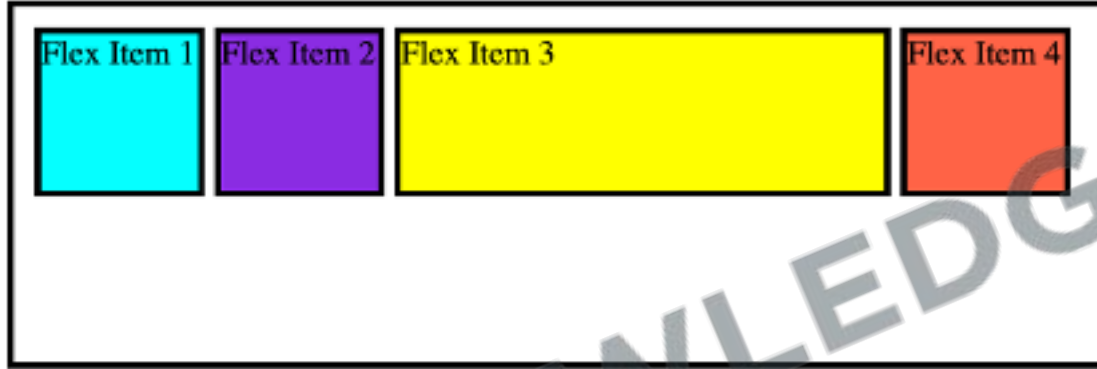
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36. Properties: Flex Items (Flex Grow)

Flexbox Container



```
#box3 {  
  background-color: yellow;  
  flex-grow: 1;  
}
```

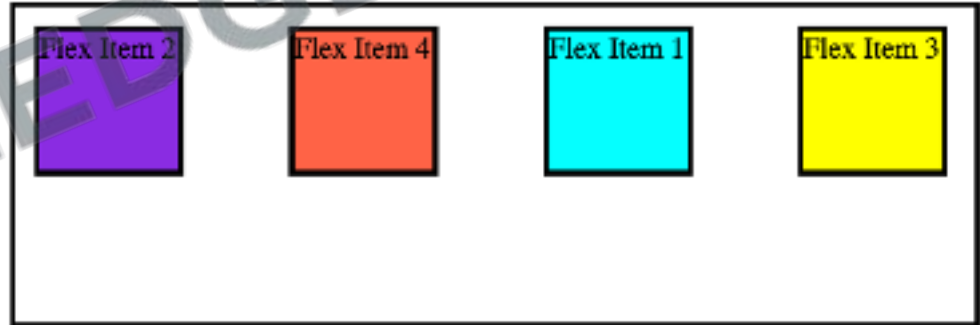
The "flex-grow" property in CSS specifies how much a **flex item** will grow relative to other items in the flex container when additional space is available.

36. Properties: Flex Items (Order)

```
#box1 {  
  background-color: ■ aqua;  
  order: 3;  
}  
#box2 {  
  background-color: ■ blueviolet;  
  order: 1;  
}  
#box3 {  
  background-color: ■ yellow;  
  order: 4;  
}  
#box4 {  
  background-color: ■ tomato;  
  order: 2;  
}
```

The "order" property in CSS allows you to define the sequence in which flex items appear within the flex container, **overriding their original order** in the HTML.

Flexbox Container



37. Grid

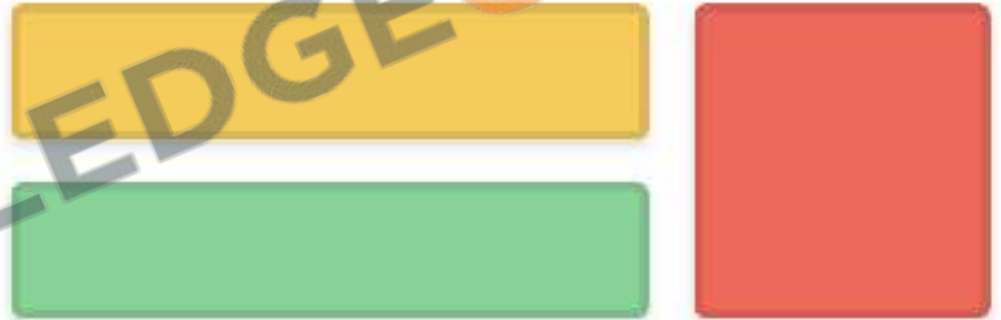
Flexbox

One-dimensional layout



Grid

Multi-dimensional layout



37. Grid

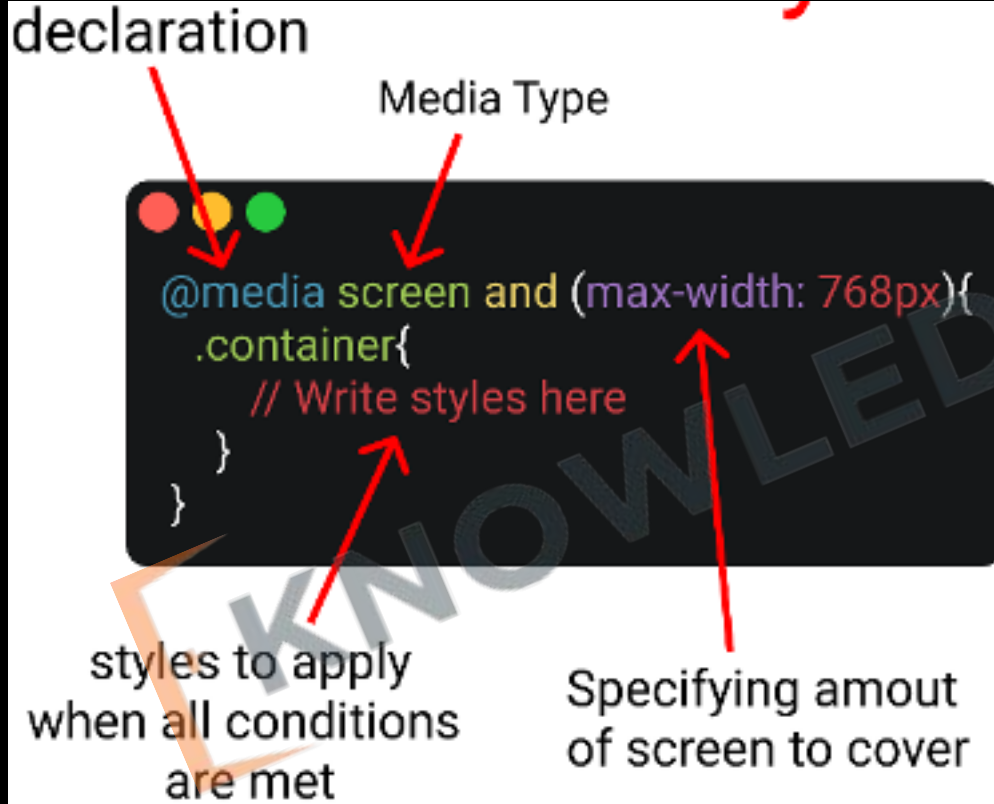
```
.container {  
  display: grid;  
  grid-template-columns: 50px 50px;  
  grid-template-rows: 50px 50px;  
}  
  
.item1 {  
  grid-column: 1 / 2;  
  grid-row: 1 / 2;  
  background-color: lightblue;  
}  
  
.item2 {  
  grid-column: 2 / 2;  
  grid-row: 1 / 2;  
  background-color: lightgreen;  
}  
  
.item3 {  
  grid-column: 1 / 2;  
  grid-row: 2 / 2;  
  background-color: lightpink;  
}  
  
.item4 {  
  grid-column: 2 / 2;  
  grid-row: 2 / 2;  
  background-color: lightyellow;  
}
```

```
<div class="container">  
  <div class="item1">Item 1</div>  
  <div class="item2">Item 2</div>  
  <div class="item3">Item 3</div>  
  <div class="item4">Item 4</div>  
</div>
```



- 2D layout system for rows & columns.
- Activate with display: grid;
- Children become grid items.
- Define structure with grid-template properties.
- Individual units called grid cells.

38. Media Queries



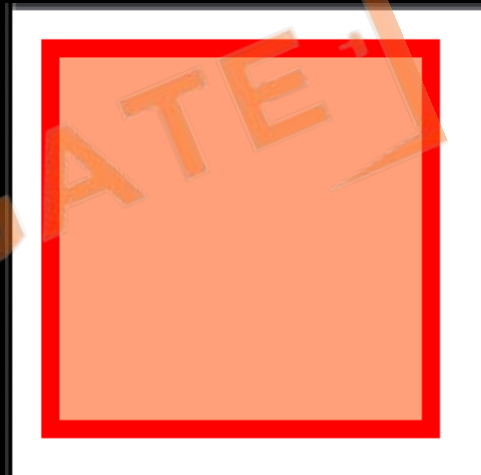
- Tailor styles for specific device characteristics.
- Use to create **responsive** web designs.
- Apply styles based on conditions like screen size.
- Syntax: **@media (condition) { CSS rules }**.
- Can **combine** multiple conditions using **and**, **or**.

38. Media Queries (width)

```
.box {  
  height: 100px;  
  width: 100px;  
  background-color:  lightsalmon;  
}
```

```
@media screen and (width: 250px) {
```

```
  .box {  
    border: 5px solid  red;  
  }  
}
```



38. Media Queries (min-width)

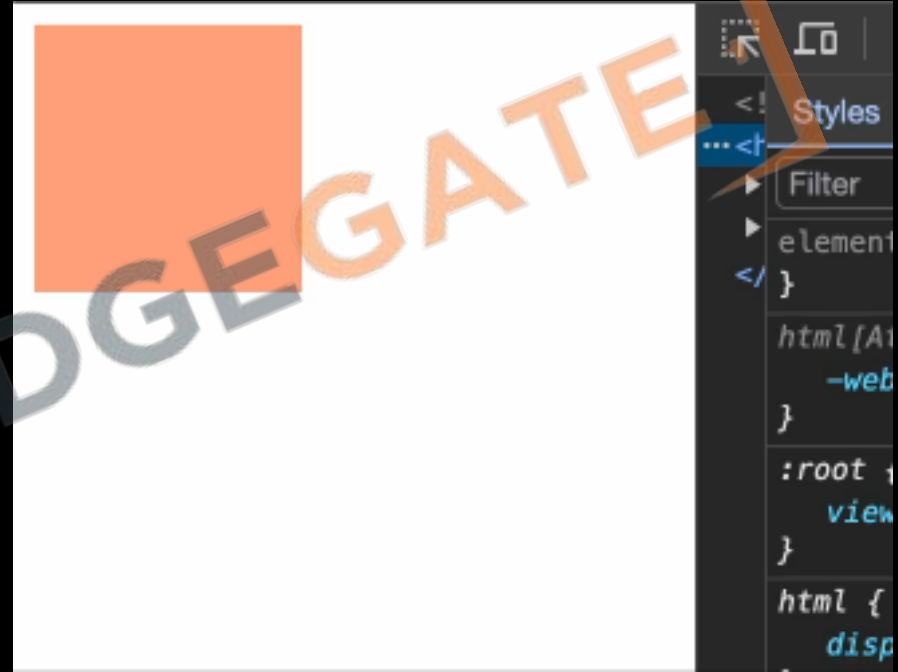
```
.box {  
  height: 100px;  
  width: 100px;  
  background-color: lightsalmon;  
}  
  
@media screen and (min-width: 300px) {  
  .box {  
    height: 150px;  
    width: 150px;  
  }  
}
```



38. Media Queries (max-width)

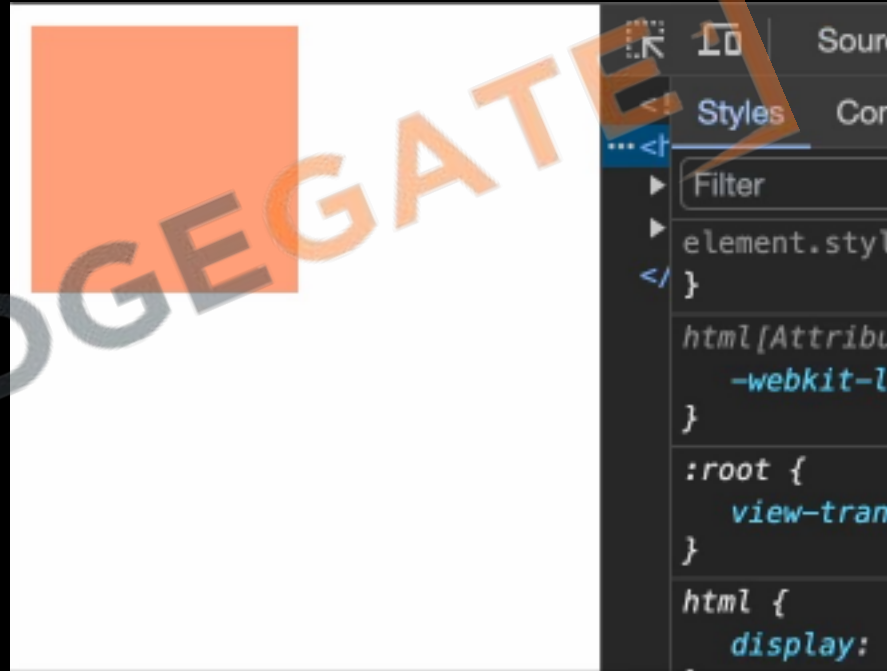
```
.box {
  height: 100px;
  width: 100px;
  background-color: lightsalmon;
}

@media screen and (max-width: 250px) {
  .box {
    height: 50px;
    width: 50px;
  }
}
```



38. Media Queries (combination)

```
.box {  
  height: 100px;  
  width: 100px;  
  background-color: lightsalmon;  
}  
  
@media screen and (min-width: 250px)  
and (max-width: 300px) {  
  .box {  
    border-radius: 50%;  
  }  
}
```



Level 6 Revision

Flex Box, Grid and Media Queries

- 31. Float Property
- 32. What is Flexbox?
- 33. Flex Model
- 34. Flexbox Direction
- 35. Properties: Flexbox container
- 36. Properties: Flex Items
- 37. Grid Layout
- 38. Media Queries



Practice Set Level 6

Flex Box, Grid and Media Queries

- Create the same nav bar created in the last Practice Set but do the spacing using flexbox.
- Use flexbox to centre an image inside a div.
- Create a div with three boxes where two boxes have fixed size, but the third box grows and shrinks with container size.
- Create a div of 50px by 50px with color green. It should respond to following conditions:
 - Keep color green under 300px of view port size
 - Change color to red from 300px to 400px
 - Change color to blue after 400px



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Level 7

Animation, Transition & Transform

- 39. Pseudo Classes
- 40. Transitions
- 41. CSS Transform
- 42. Animation

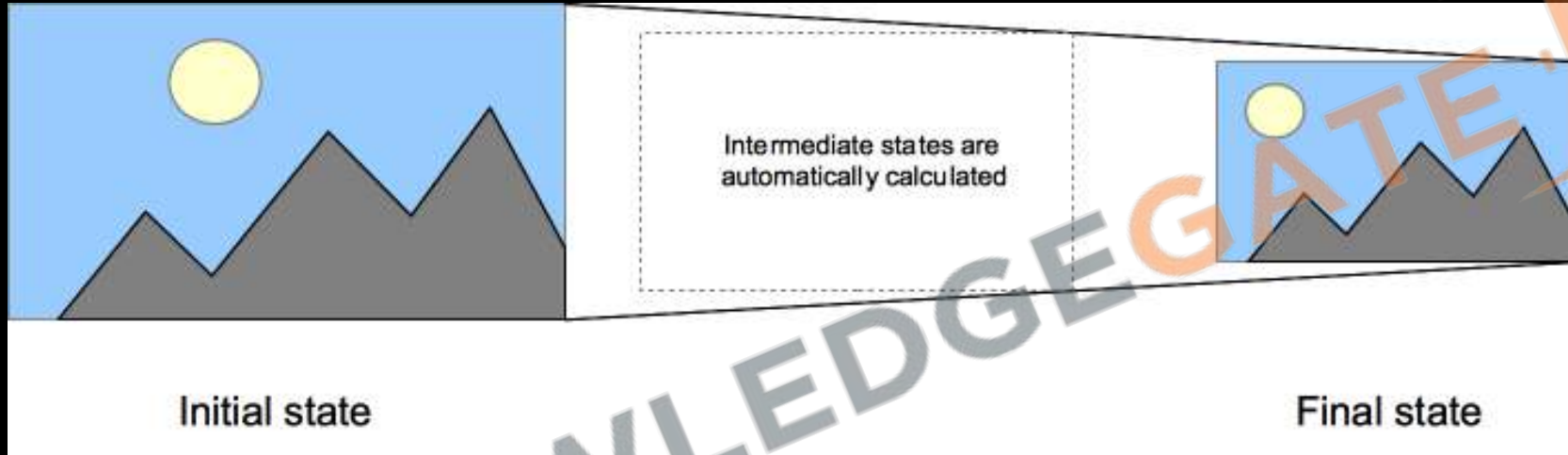
39. Pseudo Classes

```
.btn {  
    height: 20px;  
    width: 70px;  
    border: 1px solid blue;  
    border-radius: 5px;  
    background-color: aliceblue;  
}  
  
.btn:hover {  
    height: 25px;  
    width: 80px;  
    border: 1px solid red;  
}  
  
.btn:active {  
    height: 25px;  
    width: 80px;  
    border: 1px solid red;  
    background-color: indianred;  
}
```

- Used to define special states of HTML elements.
- Syntax: `selector:pseudo-class { styles }`.
- Common examples: `:hover`, `:active`, `:first-child`.
- Target elements based on their position or user action.



40. Transitions



CSS transition is a property that enables smooth animation between changes in CSS property values

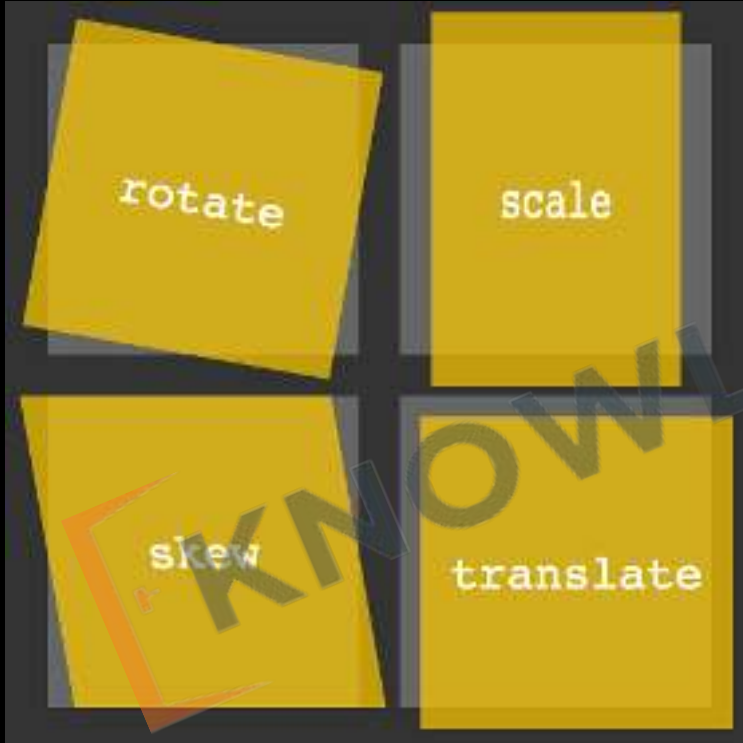
- **transition-property**: Defines which CSS properties will transition.
- **transition-duration**: Sets how long the transition lasts.
- **transition-timing-function**: Controls the speed curve of the transition.
- **transition-delay**: Specifies a delay before the transition starts.

40. Transitions

```
.btn {  
  height: 20px;  
  width: 70px;  
  border: 1px solid blue;  
  border-radius: 5px;  
  background-color: aliceblue;  
  transition-property: all;  
  transition-duration: 1s;  
  transition-timing-function: ease-in-out;  
  transition-delay: 1s;  
  /*transition: all 1s ease-in-out 1s;*/  
}  
  
.btn:hover {  
  height: 25px;  
  width: 80px;  
  border: 1px solid red;  
}  
  
.btn:active {  
  height: 25px;  
  width: 80px;  
  border: 1px solid red;  
  background-color: indianred;  
}
```



41. CSS Transform



- Allows modification of an element's shape and position.
- Can perform operations like **rotate**, **scale**, and **translate**.
- Does not affect the layout of surrounding elements.
- Used to create visual effects like **3D** space transformations.
- Implemented with functions like **rotate()**, **scale()**, and **translate()**.

41. CSS Transform (Rotate)

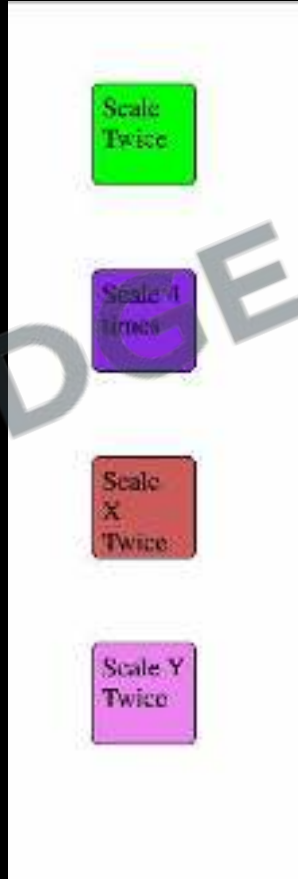
```
.box {  
  height: 50px;  
  width: 50px;  
  padding: 5px;  
  margin: 20px;  
  border: 1px solid black;  
  border-radius: 5px;  
  transition-property: all;  
  transition-duration: 1s;  
  transition-timing-function: ease-in-out;  
}  
#box1 { background-color: lime;}  
#box1:hover { transform: rotate(45deg); }  
#box2 { background-color: blueviolet;}  
#box2:hover { transform: rotate(180deg); }  
#box3 { background-color: indianred;}  
#box3:hover { transform: rotate(45deg); }  
#box4 { background-color: violet;}  
#box4:hover { transform: rotate(45deg); }
```



- Rotates an element around a fixed point.
- Defined using the rotate() function within the transform property.
- Default rotation point is the element's center.

41. CSS Transform (Scale)

```
.box {  
  height: 50px;  
  width: 50px;  
  padding: 5px;  
  margin: 50px;  
  border: 1px solid black;  
  border-radius: 5px;  
  transition-property: all;  
  transition-duration: 1s;  
  transition-timing-function: ease-in-out;  
}  
#box1 { background-color: lime;}  
#box1:hover { transform: scale(2); }  
#box2 { background-color: blueviolet;}  
#box2:hover { transform: scale(4); }  
#box3 { background-color: indianred;}  
#box3:hover { transform: scalex(2); }  
#box4 { background-color: violet;}  
#box4:hover { transform: scaley(3); }
```



41. CSS Transform (Translate)

```
.box {  
  height: 75px;  
  width: 75px;  
  padding: 5px;  
  margin: 50px;  
  border: 1px solid black;  
  border-radius: 5px;  
  transition-property: all;  
  transition-duration: 1s;  
  transition-timing-function: ease-in-out;  
}  
#box1 { background-color: lime;}  
#box1:hover { transform: translate(50px); }  
#box2 { background-color: blueviolet;}  
#box2:hover { transform: translate(50px, 50px); }  
#box3 { background-color: indianred;}  
#box3:hover { transform: translateX(50px); }  
#box4 { background-color: violet;}  
#box4:hover { transform: translateY(50px); }
```



41. CSS Transform (Skew)

```
.box {  
  display: inline-block;  
  height: 75px;  
  width: 75px;  
  padding: 5px;  
  margin: 50px;  
  border: 1px solid black;  
  border-radius: 5px;  
  transition-property: all;  
  transition-duration: 1s;  
  transition-timing-function: ease-in-out;  
}  
#box1 { background-color: lime;}  
#box1:hover { transform: skew(45deg); }  
#box2 { background-color: blueviolet;}  
#box2:hover { transform: skew(90deg); }
```



42. Animation

The keyframes at rule rule start with "@keyframes" keyword

animation name which is specified in animation-name property

```
@keyframes animation-name {  
  from {  
    /* CSS code */  
  }  
  to {  
    /* CSS code */  
  }  
}
```

specify where animation should end.
You can write 100% as well instead of "to"

Specify when the style change will happen.
You can write 0% as well, which is same as "from"

CSS properties.

42. Animation Properties

- **animation-name:** Specifies the name of the @keyframes defined animation.
- **animation-duration:** Defines the total time the animation takes to complete one cycle.
- **animation-timing-function:** Controls the pacing of the animation (e.g., linear, ease-in).
- **animation-delay:** Sets a delay before the animation starts, allowing for a pause before initiation.
- **animation-iteration-count:** Indicates the number of times the animation should repeat.
- **animation-direction:** Specifies the direction of the animation, allowing for reverse or alternate cycles.

42. Animation

```
.box {  
  height: 75px;  
  width: 75px;  
  border: 1px solid black;  
  border-radius: 5px;  
  position: absolute;  
  left: 10px;  
  background-color: lime;  
  
  animation-name: ghumakkad;  
  animation-duration: 4s;  
  animation-timing-function: ease-in-out;  
  animation-delay: 0s;  
  animation-iteration-count: 4;  
  
  animation-direction: alternate;  
  
  /* animation: ghumakkad 4s ease-in-out 0s 4  
  alternate; */  
  
  @keyframes ghumakkad {  
    from {left: 10px;}  
    to {left: 300px;}  
  }  
}
```



42. Animation

Animation



```
@keyframes ghumakkad {  
  0% {left: 10px; top: 0px}  
  50% { left: 150px; top: 100px }  
  100% {left: 300px; top: 0px}  
}
```

Level 7 Revision

Animation, Transition & Transform

- 39. Pseudo Classes
- 40. Transitions
- 41. CSS Transform
- 42. Animation



Practice Set Level 7 Revision

Animation, Transition & Transform

- Create a webpage with a progress bar that showcases a smooth loading animation. The progress bar should fill up from 0 to 100% with a smooth transition effect and a slight bounce when it reaches 100%.



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